

# POOL PLANT ROOM

AS-FOUND SURVEY



CURRENT  
VERSION

[albidr.github.io](https://albidr.github.io/Pool-ADR)  
/Pool-ADR

|                                                           |                                                |                                              |                                                            |
|-----------------------------------------------------------|------------------------------------------------|----------------------------------------------|------------------------------------------------------------|
|                                                           |                                                |                                              |                                                            |
| <b>POOL VOLUME</b><br>70 to 75 m <sup>3</sup>             | <b>POOL TYPE</b><br>Overflow +<br>balance tank | <b>POOL FINISH</b><br>Black resin,<br>smooth | <b>PLANT ROOM</b><br>4.52 m <sup>2</sup> , below<br>ground |
| <b>BALANCE TANK</b><br>5.46 m <sup>3</sup> , next<br>door | <b>FILTRATION</b><br>Sand, Kripsol 6-<br>way   | <b>PUMP</b><br>DAB EuroPro 200<br>M          | <b>SANITISATION</b><br>Salt, Zodiac TRi<br>PRO             |
| <b>FILTER</b><br>Kripsol Ø760,<br>225 kg                  |                                                |                                              |                                                            |

## QUICK REFERENCE

### EVERYTHING NORMAL

- (M) on (1), Filtration
- (B) and (C) open, (A) almost closed
- All four (D) open
- (G) open (it is also (7)), (E) open
- (L) shut or barely open, (H) shut, (1) shut
- Column **ISOLATOR** open, and the column moving
- Gauge (4) at **about 1.18 bar**, green on the dial
- pH 7.2 to 7.4, chlorine 0.6 to 1.5

### WATER COMING IN

- The pool is above you. **Stopping the pump does not stop the water.**
- Shut (7), then (L), then all four (D), then (G)
- Check the sump pump (N) is running. It is the only drain.

### THREE THINGS NEVER TO DO

- Never move the Kripsol lever with the pump running. Kill the panel first.
- Never put a chlorine product near the acid drum. The drum is **sulfuric acid**, whatever the old sheet says.
- Never run the cell with no flow. Isolate on (B) and (C).

## WHERE EACH STATEMENT COMES FROM

### OBSERVED

Read off the photographs: nameplates, printed labels, and pipe runs I could follow end to end. Counted against a second image wherever countable.

### MANUFACTURER DOCUMENTATION

The makers' own manuals for the filter, the chlorinator, the TRi PRO module and the pump, downloaded and kept in *10\_manuals/* with the official link for each. Where a manual and the printed sheets disagree, this document follows the manual and says so.

### HOUSEHOLD REFERENCE

The four printed sheets kept in the room, written for this pool and used for years. Not a professional specification, and not necessarily complete. They are the standing house procedure and the shared vocabulary for this plant room, and they are the primary record of how it was run.

### INFERRED

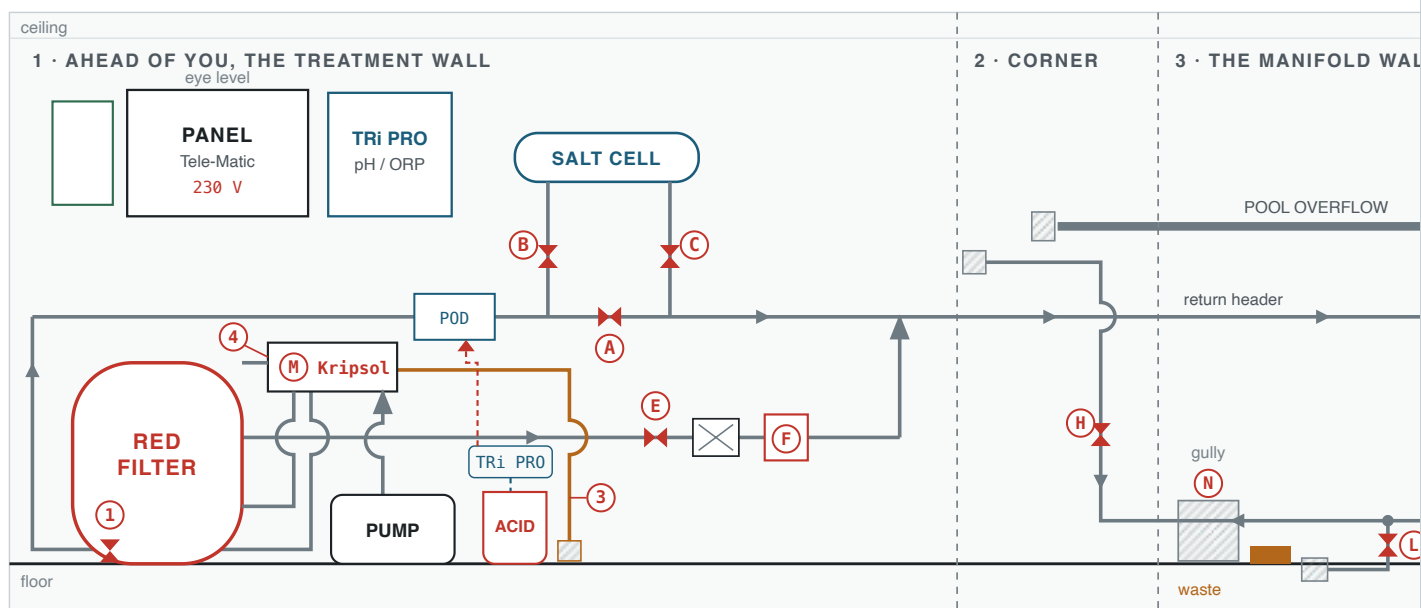
My reading rather than an observation. Very little is left: what remains is named as mine where it appears, most of all the isolation sequence in the safety notes, which follows from the levels and not from the sheets.

## §1The whole installation, one diagram

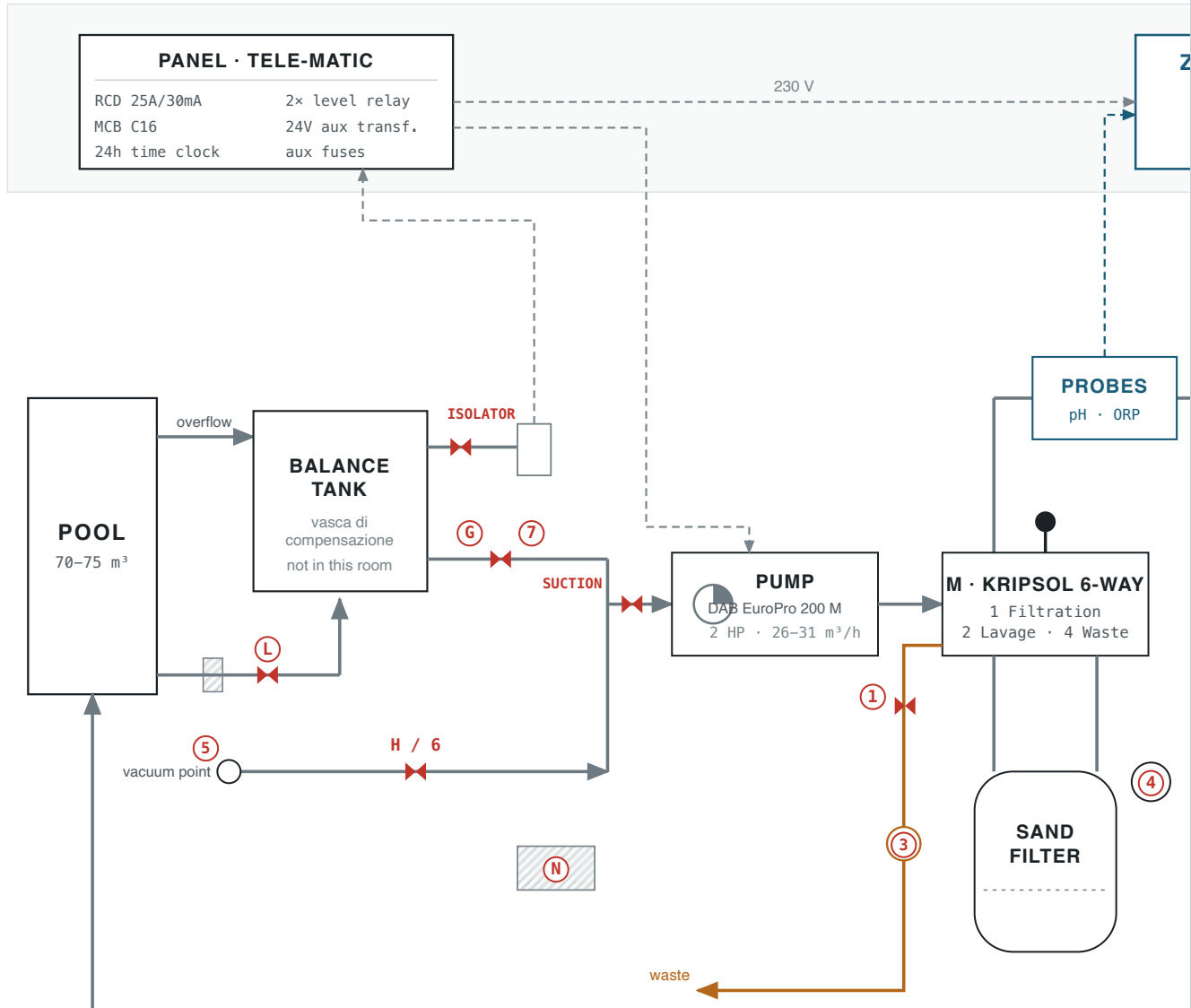
This uses the letters and numbers already written on the printed sheets ((A), (B/C), (D), (E), (F), (G), (H), (L), (M), (N), and (1) to (7)) rather than tags of my own, so that the drawing, the sheets and anything said out loud in the room all use one vocabulary. Every key is drawn as a circle, on the drawing and in the text alike.



STAND IN THE DOORWAY, TURN RIGHT, AND THIS IS WHAT YOU PASS

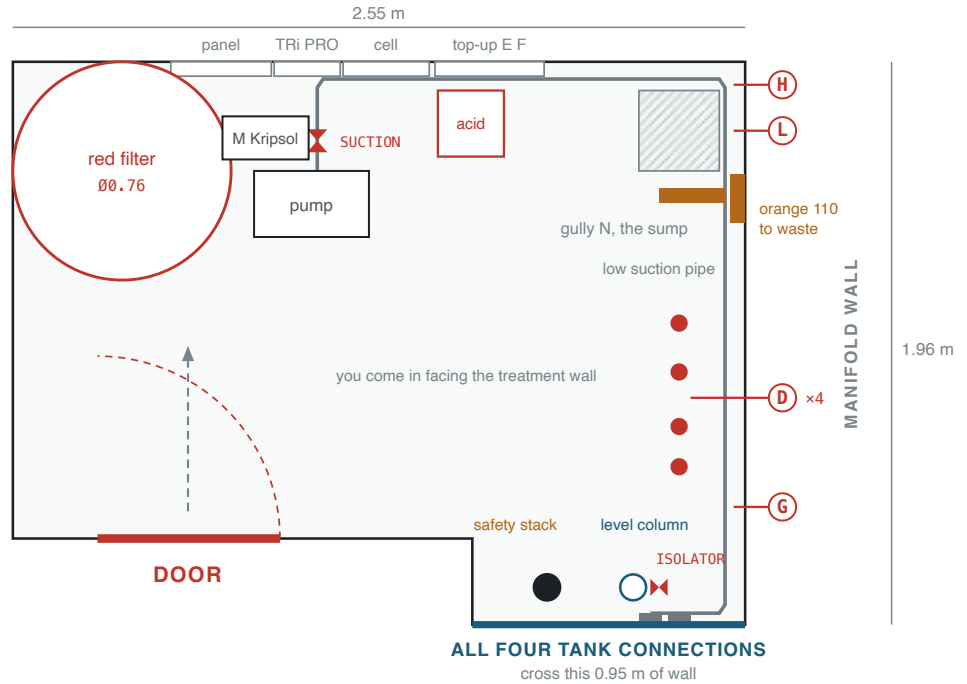


## ELECTRICAL, DOSING & CONTROL



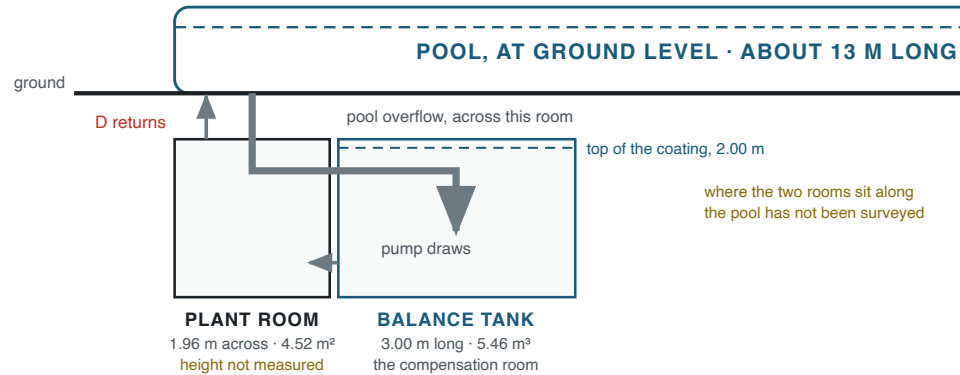
### PLAN OF THE PLANT ROOM

to scale, 1 m = 200 units · 4.52 m<sup>2</sup> of floor · positions off the photosphere and the four reprojections, corrected on site



### SECTION: HOW THE SPACES STACK UP

to scale sideways, 1 m = 55 units, the same as the site plan · heights are indicative



**Five views, and two of them are now to scale.** The wall strip is what you see turning right from the doorway; its order follows the 360° pan and its distances are not to scale. **The plan of the plant room is to scale** at 1 m = 200 units, drawn from your measured 2.55 by 1.96 m with the 1.60 by 0.30 m entrance notch, so 4.52 m<sup>2</sup> of floor. **The site plan is to scale for the rooms** at 1 m = 55 units: the balance tank runs out at right angles from the bottom-right corner, 0.91 by 3.00 m, coated 2.00 m up, and all four of its connections cross that one 0.95 m of wall. The service corridor on it is schematic, because its size has not been measured. The section shows the pool sitting above both underground rooms, which is why its water will find any opening down here. Every tag is explained below.

## READING THE DRAWING

- pipe    — larger bore    — to waste    ----- electrical or signal    ◀▶ a valve
- ⤿ a small arc means two pipes cross without joining    — a wall the balance tank is behind
- Ⓔ Ⓕ two badges on one valve: the sheets give it two names
- on the site plan, schematic and not measured

## THE LETTERS, AS THE SHEETS NUMBER THEM

- Ⓐ Bypass around the salt cell. **Keep almost closed:** it sets how much water skips the cell.
- Ⓑ, Ⓒ Cell inlet and outlet. **Keep open.**
- Ⓓ ×4 Return valves on the manifold. **All four stay open**, or the circuit throttles and filter pressure rises.
- Ⓔ Mains top-up tap. **Keep open** so the automatic top-up can work. The line runs on to the left behind the filter and **discharges into the return manifold**, so mains water reaches the pool through the four wall outlets. It does not feed the tank; the tank fills from the pool spilling over its deck channel.
- Ⓕ Solenoid on the top-up, opened by the two level relays in the panel. A manual bypass loop beside it lets you fill by hand if it fails, and it is also the way to speed up a refill, because a top-up solenoid is a small orifice by design. **The bypass has no cut-out**, so close it before the pool crests or it will run on into the tank's safety overflow and out to waste.
- Ⓖ Compensation valve, on the low pipe behind the return drops. It is the tank outlet's own valve, and therefore **the pump's entire water supply. It stays fully open.** It is not the valve carrying the "barely open" advice; that is Ⓕ, below, and the two are easy to confuse. With Ⓕ and Ⓗ shut, which is normal running, every litre the pump moves passes through Ⓖ. Two things follow. **Never throttle it.** A restriction before the pump drops the pressure at its inlet and can boil the water there, which sounds like gravel in the casing and erodes the impeller; if flow ever has to be trimmed it is trimmed after the pump, never before. And **never run the pump with Ⓖ and Ⓕ both shut.** That is a dry run, and it is the one the low-level interlock cannot catch: the interlock watches the tank's *level*, so the tank can be brim full and the clear column perfect while the path to the pump is closed. The only reason to shut Ⓖ is emptying the pool, where Ⓕ is opened in the same operation, with the pump off, so the pump still has somewhere to draw from.
- Ⓗ Also Ⓔ on the cleaning sheet. Manual vacuum. **Normally shut.** The port is in the pool wall: unscrew the cover, push the hose in, and the pump pulls dirt off the floor.
- Ⓕ Pool floor suction, the *presa di fondo*. **Shut, or barely open** so the deep water is renewed rather than sitting still. Open fully to empty the pool, with the pump off and Ⓖ shut in the same operation. **The pump must always have exactly one suction path open**, and normally that is Ⓖ.



Ⓜ Also ② on the cleaning sheet, which is a picture number and not a lever position. Kripsol six-way valve, mounted on the filter's side. **Position ①, Filtration** for normal running.

**POD** The clear probe holder clamped to the low horizontal pipe, between the filter and the cell branch. One body doing four jobs: the **flow switch** that stops the cell when the water is not moving, the **pH probe** (red), the **ORP probe** (blue), and the **acid injection point**. Zodiac's name for it, and the reason everything measured or dosed happens in one place.

**ACID** Sulfuric acid 36%, the pH minus. One tube runs from the drum to the TRi PRO's peristaltic pump; a second returns from the pump to the **POD**, the clear probe holder on the low horizontal pipe. That is where it injects, into the same body that carries the pH probe, the ORP probe and the flow switch. The POD sits **before** the cell, which is where Zodiac puts it: the manual specifies the POD as the first element after the filter.

**ISOLATOR** The valve at the foot of the level column, on the take-off between the **balance tank and the column**. Red handle, and the only valve in this room the printed sheets never lettered, which is why it is named here rather than given a letter of my own. **Keep it open**. Shut, it cuts the column off from the tank, and everything that follows is silent: the column freezes at whatever level it was showing, the floats sitting in it read that stale level, the automatic top-up stops calling for water, and **the low-level interlock reads a satisfied tank and will let the pump run dry**. Nothing else in the room looks wrong. If the column has not moved in days, check this before believing it.

Ⓝ Floor gully and sump with a submersible pump. The room is below ground, so this is the only way water leaves the floor.

① Filter drain valve. Open it before a backwash, close it after.

③ Sight glass. Run the backwash until the water in it comes clear.

④ Filter pressure gauge. **About 1.18 bar is normal running on this installation, measured 2026-09-09 on clean sand**. Its face is banded green to 1.48 and red beyond, so the wash trigger is printed on the instrument. Kripsol's manual gives 0.5 to 0.7 and a 1.0 backwash trigger, but those describe the filter alone; this one sits behind an oversized pump on limiting pipework with the cell in circuit and ⓐ throttled, and all of that adds head. The printed sheet's "about 1 bar" was the accurate figure for this room. **Backwash on the climb away from 1.2, not on an absolute number**, and see the note on headroom in the water figures.

⑤ The vacuum point itself, the socket in the pool wall: unscrew the cover and push the hose in. A picture number from the cleaning sheet rather than a valve. The valve that opens that line is ⑥, which is ⓗ.

**SUCTION** The red-handled ball valve on the pump's suction, immediately before the clear strainer. No letter on sheet 1 and no picture number on sheet 3, so it is named here rather than given a letter of my own. **Keep it open**. It is what you shut to open the strainer basket or work on the pump without draining the balance tank, and it is the "suction valve" Kripsol's pre-start check tells you to open before filling the pump body by hand. Visible in photographs 16, 25 and 26. **No matching valve is visible on the discharge in any photograph**, so where Kripsol says "the suction and discharge valves", this installation appears to have one, not two.

⑦ **The same valve as ⑥.** Balance tank outlet: close it to vacuum harder, and close it when emptying so the tank does not drain too. This was carried as unresolved until the printed sheets were read against each other. *Sheet 1* letters the components on its own photograph of the room, ① to ⑨, ①, ②, ③, ④, ⑤. *Sheet 3* carries its own strip of seven numbered photographs and cites components by picture number, ① to ⑦. Three components appear on both and so carry both labels. Sheet 3 gives itself away: its backwash note says **close valve ⑥** because the tank level drops, and its vacuum step says **close the balance tank outlet ⑦**, which is the same valve, the same action and the same sheet. Picture 7 shows a red-handled ball valve on an elbow, which is what stands at the tank end of the low run.

#### PANEL SETTINGS FOR NORMAL RUNNING

**pump** Filtration pump on ①, automatic, so the **Hager EH011** time switch marked **PROG. POMPA** runs it. **Read the clock before you trust ①.** On 2026-09-10 its index sat near 22:00 in a photograph timed 01:47, about four hours slow. It is a synchronous motor clock: it stops dead whenever the panel is switched off, never catches up, and the panel is switched off for every Kripsol lever move. Nothing about a slow clock looks wrong from across the room. It simply runs the pump at hours nobody chose.

**top-up** Reintegro on ①, automatic, so the level relays call for water on their own and shut it off when the tank is satisfied. [*settled 2026-09-10*] Photograph 35 shows a **Lovato LVM 20** level relay for conductive liquids carrying the legend **CONTROLLO LIVELLO REINTEGRO**, so the top-up's automatic control is a level relay and not a timer, which this document had left open. What is still unknown is where the probes sit and at what levels they are set. On ③ that relay is bypassed and there is no cut-out at all. The top-up discharges into the **return manifold**, so mains water reaches the pool through the four wall outlets, not into the tank.

**interlock** **There is a low-level cut-out that will not let the pump run when the balance tank is too low.** The printed sheet states it plainly in the backwash procedure, "the balance tank level drops during this and will stop the pump", and it was confirmed by accident on 2026-09-07 when the pump sat on ③ for a full day with an empty tank and never started. Treat it as a backstop for the day someone forgets, never as a substitute for the selector being on ⑧: the same clear column that feeds it has its own **ISOLATOR**, and closed, that valve makes the floats read a satisfied level and the interlock **permit** a dry run.

#### HOW THE WATER MOVES, IN ORDER

1. The pool spills over its deck channel into the **balance tank**, which is in the compensation room next door, not in here. The big pipe crossing the manifold wall high up is that overflow passing through. The pool is surrounded by a **wooden deck** and the water passes beneath it. The overflow is an **infinity edge on one side only, the long side of roughly 12 m**. A weir that long pins the pool surface to within about a millimetre of the crest at any realistic flow, so once the pool is spilling, essentially every litre arriving in the pool goes straight on to the tank.
2. The pump draws from the balance tank through ⑥, which the cleaning sheet calls ⑦. The **pool floor suction ①** and the **manual vacuum ②** feed the same low pipe, which runs behind the four return drops

and crosses them without joining.

3. The pump pushes up into the Kripsol valve (M), which on position (1) sends the water into the **filter's upper union**.
4. It comes back out of the **filter's lower union** through (M), drops to the floor, runs left behind the filter, up the wall, and rejoins the main run by the panel.
5. From there it passes the **probes** and the **salt cell**, with (A) setting how much skips the cell.
6. The **return header** takes it along the manifold wall and down the four (D) drops, through the floor, back to the pool.
7. On backwash, waste leaves through (M) along the floor and out through the wall in the **orange 110**.

#### NOT IN THIS ROOM

**tank** The balance tank runs out at right angles from the bottom-right corner of this room: 0.91 m wide, 3.00 m long, coated 2.00 m up, so 5.46 m<sup>3</sup> to the coating line. Its four connections come through that corner wall. Reached only by a small hatch at the far end, 3 m from the connections. Photographs in *05\_balance-room/*, dimensions in *DIMENSIONS.txt*.

**valve** The main shut-off for the whole pool system is in the boiler room.

§2The two rooms, measured

Measured 2026-09-07, so the drawing above and the tape now agree. The dimensioned sketch and the scaled site plan are in 07\_drawings/; the full working is in DIMENSIONS.txt.

|            | THIS PLANT ROOM                                                                                                           | THE BALANCE TANK                                                                                                      |
|------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Shape      | 2.55 × 1.96 m, with a 1.60 × 0.30 m notch at the entrance corner                                                          | 0.91 × 3.00 m, coated 2.00 m up                                                                                       |
| Floor      | 4.52 m²                                                                                                                   | 2.73 m²                                                                                                               |
| Volume     | At least 2.33 m tall from the 3D scan, so <b>10.5 m³ of air or more</b> . Not closed: the scan captured no ceiling plane. | 5.46 m³ to the coating line, but the safety overflow is at 1.35 m, so <b>3.69 m³ is the most it ever holds</b>        |
| Getting in | The door, bottom left on the plan.                                                                                        | One small hatch high in the service corridor. A ladder to climb to it and a second lowered inside to reach the floor. |

THE TANK'S FOUR CONNECTIONS, ALL THROUGH ONE 0.95 M OF WALL

- in

The pool overflow. A large orange-brown pipe entering high, ending open and **lying loose on the tank floor**. Clipping it would cost nothing and stop it knocking when the pool spills hard.
- out

The tank outlet, **0.35 m above the tank floor**, through the compensation valve **G**. **This is the pump's entire water supply**. On an overflow pool the pump does not draw from the pool, it draws from the tank. Below 0.35 m the pump has nothing; start it at about 0.55 m so the socket has cover and does not vortex.
- level

The take-off for the clear column in this room, alongside the outlet at **0.35 m**, so the column can only read levels above that. Communicating vessels: the column stands at exactly the tank's level, so it is a remote gauge for a room you cannot see into. Its float switches drive the two level relays, which open solenoid **F**. **An isolator sits on this take-off at the column's foot**, so the one line carrying the tank's level into this room can be shut by hand; see ISOLATOR in the diagram legend for what that costs.
- safety

The high-level overflow to waste, **1.35 m above the tank floor**, not at the top: there is 0.65 m of coated tank above it that never holds water. No valve, no electrics, nothing to fail. If the tank fills past it, water leaves instead of rising.

FOUR NUMBERS WORTH CARRYING

- 27.3 L

per centimetre of tank level. This one turns the clear column from a vague indicator into a measuring instrument.

- 5.1%** of pool volume, taking the 3.69 m<sup>3</sup> the tank can actually reach. Design practice for a deck-level pool starts at 5 to 10 percent, so **the tank is adequate rather than generous**: it sits at the floor of that band, and the working band alone is 3.8 percent, below it. The 7.5 percent you get from the coating line is not reachable in normal operation.
- 29 cm** of level is what ten people getting in at once takes out, at the design figure of 80 litres a bather. Bather surge is not this tank's constraint.
- 33–55 cm** **is what one backwash and rinse takes**, out of a band that is exactly 1.00 m. That is the constraint, and it is why the sheet warns the pump will stop. Ten bathers take another 29 cm, so **do not backwash with a full pool of people in it**.

#### WHY THE SHEET SAYS THE PUMP WILL STOP DURING A BACKWASH

- Two to three minutes of backwash at 18 to 26 m<sup>3</sup>/h, plus the rinse, removes roughly **900 to 1,500 litres**. Against a working band of about 2,700 litres that is **a third to a half of everything the tank has to give, in one cycle**.
- So the note on the printed sheet is not vague caution. It is an accurate description of a tight margin. **Top the tank up before you backwash, not after**.
- That same water carries salt away with it: 3.6 to 6.0 kg. One 25 kg sack replaces the loss of four to seven backwashes.

#### BEFORE ANYONE GOES INTO THE TANK

- It is a confined space. One small opening, high up, below ground, no second way out, ladder in and ladder down. **Never alone**.
- The four connections are on the wall **3 m from the hatch**, so reaching them means traversing the full length of a chamber 0.91 m wide and 2.00 m deep.
- The lowest connection is at 0.35 m and there is no sump, so **the tank cannot be emptied through any fixed connection: 956 litres** stay below the outlet and have to be pumped or sponged out.
- *[added]* There is rust-coloured staining running down the corridor wall below the hatch, and blown plaster at the sill. **Read this as damp, not as a leak**. That corridor is an **intercapedine**, a cavity deliberately left around the below-ground structure to keep moisture away from it, so it is damp by design and doing its job. A below-ground tank of water on the other side of the wall does not help. The question worth answering is not where the water is getting in, but **whether the intercapedine is actually ventilated**, because a sealed one stops working, and separately what the rust is running off.

### §3Procedures, as written on the sheets

Transcribed and translated from the four printed sheets, with their numbering kept. I have not changed the order of any step. Where I have added something the sheets do not say, it is marked *[added]*.

BEFORE MOVING THE KRIPSOL LEVER

- The sheets are explicit: **switch the panel off at the main switch first**, move the lever, then switch back on. Turning that valve against a running pump is what destroys the seal inside it.
- [added]* Never run the pump with the lever between positions.
- [added]* During backwash, rinse and waste the water goes to drain rather than back to the pool, so the cell should not be producing into the sewer.

| KRIPSOL POSITION | MARKED ON THE DIAL        | WHAT IT DOES                                                                                  |
|------------------|---------------------------|-----------------------------------------------------------------------------------------------|
| ①                | FILTRACION / FILTRATION   | Normal running. Down through the sand, out to the returns.                                    |
| ②                | LAVADO / BACKWASH         | Reverse flow, up through the sand, out to waste.                                              |
| ③                | ENJUAGUE / RINSE          | Settles the bed after a backwash, still to waste. This is the "fare risciacquo" step.         |
| ④                | DESAGUE / WASTE           | Straight to drain, bypassing the filter. Used for emptying and for vacuuming heavy dirt away. |
| ⑤                | CIRCULACION / RECIRCULATE | Round the circuit with the filter bypassed.                                                   |
| ⑥                | CERRADO / CLOSED          | Shut. Never run the pump on this.                                                             |
| ⑦                | INVIERNO / WINTER         | Winter position, handle parked between seats so the gasket is not compressed all season.      |

Cast around the rim of the valve itself, in four languages: **NEVER SHIFT HANDLE WHILE PUMP IS RUNNING.**

WHAT THE FILTER IS DOING, AND WHEN IT NEEDS WASHING

the point

The filter catches dirt and **keeps** it, inside the sand. Eventually the sand is full and catches nothing more. **A backwash is how you empty it:** the flow is reversed so the water travels up through the bed instead

of down, which shakes the dirt loose, and that dirty water goes to the drain rather than back to the pool. Emptying a vacuum cleaner, flushed out with water instead of binned.

### not a sieve

The gaps between sand grains are far bigger than the particles the bed catches. It works by **contact**: water threads through a maze of pore spaces and particles collide with grain surfaces on the way and stick. Time in the maze is everything, which is why **velocity is this filter's weakness** - push water through fast enough and fine particles are carried past before they can attach. It is also why a flocculant works: it does not make the sand finer, it makes the **particles bigger**, so they cannot get through.

### why the gauge rises

Dirt fills the pore spaces, mostly in the top few centimetres. The same flow through narrower passages meets more resistance, and gauge ④ sits upstream of the bed, so it reads that resistance directly. **A rising gauge is the filter reporting that it is filling, not a fault. A falling gauge with a noisy pump is the opposite problem and a far more urgent one: gauge 4 sits after the pump, so a starved suction reads low, not high. High and quiet means the bed. Low and rattling means the suction, and that is ⑥, the strainer basket, or the tank.**

### when to wash it

**On the gauge, never on a schedule. 1.18** is a clean bed with nothing to wash out. **1.3 to 1.4** is loading and working properly, leave it. **Around 1.5** is full: it will catch no more until washed. **1.6** is the vessel's plate rating, so never get there. A label telling you to backwash every few hours is written for someone clearing water in a hurry; going by the gauge uses less water and reaches the same place. **You do not have to remember any of these numbers at the gauge itself.** Its face is banded green and red, and the break was measured off photograph 37 at **1.48 bar**. Green means leave it, red means wash it, and the instrument agrees with a figure this document reached by another road entirely: Kripsol's manual, the household sheet, and a reading taken on freshly washed sand.

### the bed lifts

Above a certain upward velocity the bed **fluidises** - the grains separate and tumble, and everything trapped between them is released. That is why the vessel has empty space above the sand: the bed expands by 10 to 20 percent and needs somewhere to go. It is also why the flow must not be excessive, which would carry the sand itself out to waste.

### why the rinse

Two jobs. Fluidising left the bed loose and mixed, and the rinse sends water back down through it to **resettle** the grains. And the vessel and pipework are still full of dirty water, which the rinse flushes out **while still routed to waste**. Skip it and the first thing reaching the pool is the

dirt you just worked to remove, through a badly packed bed. Twenty to forty seconds.

**the sight glass decides**

The waste runs filthy and then clears. **Clear means the bed is clean**, and that is the endpoint rather than any number of minutes. Two to three minutes is typical. Stopping early leaves dirt in the sand; running on only wastes water.

**what it costs**

**900 to 1,500 L leave the system permanently**, which is 33 to 55 cm of the clear column, and once salt is in the water, 3.6 to 6.0 kg of salt with it. From a full tank there is room for one comfortably; consecutive backwashes need a top-up between them. Watch the column, and if it approaches the interlock, **switch the panel off**.

**and it undoes a flocculant**

A flocculant forms a fine layer on top of the sand that filters better than sand alone. **Backwashing strips that layer off with the dirt**, which is the point, but the clarifying effect goes with it. That is why the label calls for a smaller dose after every backwash: you are rebuilding the layer each time. So while a flocculant treatment is running, a backwash is not free progress - it is a reset.



## TEST THE WATER (PH AND FREE CHLORINE)

**Why it matters:** until 2026-09-10 there was no kit here and every dose went in blind. While the cell is off for want of salt, hand-dosed chlorine is the only thing keeping this water safe, and "every two or three days" is a guess dressed as a schedule. This is what turns it into a number. **Weekly in normal running, every two days while the cell is off**, and always before dosing rather than after.

1. **Sample at elbow depth, away from the four D returns and away from the wall.** Surface water is not the pool: it carries whatever is floating and has taken the most sun. Water straight off a return is what the plant just made, not what the pool holds.
2. **Rinse both cells three times with the water you are about to test**, then fill to the 10 mL mark. A cell rinsed under the garden tap reads the garden tap.
3. **One tablet per cell, dropped straight from the foil. Do not touch them.** They are hygroscopic, and skin carries both oil and chloride. **DPD No.1** for free chlorine, **phenol red** for pH.
4. Cap, and shake until dissolved, crushing the tablet against the wall of the cell if it is slow.
5. **Read the chlorine within a minute.** DPD starts fading from the moment the colour forms, so a cell left on the side while you do something else reads low.
6. Read against **daylight but not direct sun**, card behind the cell, something white behind the card.
7. **Write both numbers and the time into LOG.txt as a TEST entry.** One reading is a number; a column of them is the only thing that shows a drift before it becomes a problem.

**The trap that reads as zero.** Well above the top of this scale, DPD is bleached straight back to colourless. A cell that flushes pink and then goes clear, or never colours at all in the hour after a dose, is telling you the chlorine is very high, **not** that there is none. The test for it costs nothing: half sample, half tap water, test that, and double the answer.

**This kit stops at 3.0 mg/L, and two figures in this document are above it.** The commissioning shock is about 8 mg/L and the robot's ceiling is 4. Neither can be read directly, so use the same half-and-half dilution and double. **Test chlorine before pH** as a habit: above roughly 3 mg/L free chlorine, phenol red reads falsely high and can go purple outright, so a pH taken alongside an over-scale chlorine is not a pH at all.

## BACKWASH THE FILTER (CONTROLAVAGGIO)

Two numbering systems meet in this procedure, so read the labels not the digits. ①, ③ and ④ are components on the plan: the filter drain valve, the sight glass and the pressure gauge. ① to ⑥ after the words "lever (M)" are positions on the Kripsol. The same digit means two different things, and ③ is the worst of them: the sight glass, and Rinse.

1. **Panel off.**
2. Open filter drain valve ① (vertical).
3. Lever (M) to ②, **Lavage.**
4. **Panel on.** Run until the water in sight glass ③ runs clear. **The glass decides, not the clock;** two to three minutes is typical.
5. **Panel off.**
6. Lever (M) to ③, **Rinse. Leave drain valve ① open.** The rinse is still routed to waste and that is the whole point of it.
7. **Panel on for 20 to 40 seconds.**
8. **Panel off.**
9. Lever (M) back to ①, **Filtration.**
10. **Now** return drain valve ① to horizontal (closed).
11. **Panel on.**
12. **Read gauge ④ and write it into (LOG.txt).** That reading is the new clean baseline, and it is the number every later one is judged against. On 2026-09-09 it was 1.18.
13. If a flocculant treatment is running, **put about 72 g back in.** The backwash strips the layer off the sand along with the dirt, which is why the label asks for a smaller dose after every wash.

**The order of the last steps was wrong here until 2026-09-10,** and it is worth saying why rather than quietly fixing it. This document used to close the drain and return the lever to Filtration, and only then say "do a rinse", with no position given. Followed literally that sends the vessel's remaining dirty water straight into the pool through a bed that fluidising has just left loose and unpacked, which is precisely what the rinse exists to prevent. Rinse **before** filtration, with the drain still open.

**Note from the sheet:** the balance tank level drops during this and will stop the pump. Close valve (G) temporarily. Gauge ④ reads **about 1.18 bar** on this installation when the sand is clean. **This document does not follow the sheet on (G).** (G) is the pump's entire water supply, and closing it while the pump runs starves it, which is the one failure that destroys a water-lubricated seal in minutes. If the column falls toward the interlock, **switch the panel off instead:** same result, no risk.

## CLEAN THE PUMP STRAINER BASKET

**Why it matters:** that basket is the last thing between the pool and the impeller. Let it block and the pump is starved; let it crack and debris goes straight into the impeller, which is the expensive part. **Weekly in season**, and always after vacuuming, which is what the sheet's handwritten note is about.

1. Pump control to ⑦, **panel off**.
2. Shut ⑦ and ⑥ on the suction, and **all four ⑤** on the returns. The pool and the tank are both above the pump, so there is head from both directions. Have a towel down.
3. Unscrew the **two black knobs by hand. No tools** - they are moulded knobs on a polycarbonate lid.
4. Lift the lid, lift the basket out, rinse it with water and a soft brush.
5. **Check the basket rim for cracks.** A split basket passes everything it should be catching.
6. **Clean the O-ring and the face it seats on.** Grit there is the classic reason a pump draws air. Look for flats, nicks and hardening, and replace it if it has gone stiff rather than finding out on a dry start.
7. **Silicone grease only**, never petroleum or Vaseline: they swell the elastomer and can craze the lid.
8. Refit **hand-tight only**. Over-tightening distorts the lid, cracks the plastic around the knob bosses, and makes the seal worse rather than better.
9. Re-open ⑦ and ⑥. With the tank full the pump body **fills by gravity** - watch it come up through the lid before starting. Only prime by hand if the system is drained.

**The lid is polycarbonate**, which is attacked by solvents, alkalis and ammonia. **Water and a soft cloth only.** For limescale, a dilute citric or acetic solution is safe, then rinse well. **Never acetone, alcohol sprays, ammonia glass cleaner or an abrasive pad** - all of them craze or permanently fog it, and then you have lost the only window into the pump.

## VACUUM THE POOL FLOOR (ASPIRAZIONE FANGHI)

1. Attach the pole to the vacuum head.
2. Fill the hose with water as fully as you can, so it does not draw air.
3. Unscrew the cover and put the hose into outside point ⑤.
4. Filtration must be running, pump on.
5. Open vacuum valve ⑥ (vertical).
6. Close balance tank outlet ⑦ (transverse). It does not have to be fully closed: the more you close it the harder it sucks. If the head sticks to the floor, open it a little.
7. Reverse the process to finish.

**Handwritten on the sheet:** the basket under the clear lid after the pump has to be cleaned, by unscrewing the two black knobs. And: with the Kripsol lever on position ④, the dirt is discharged to drain instead of passing through the filter, but then the water you suck out is thrown away.

## REFILL AND RESTART, AFTER THE SYSTEM HAS BEEN DRAINED

1. **Pump control to ⑥ before anything else.** Not ①: on automatic the 24 hour time clock will start it on its own schedule, and the pump's seal is water-lubricated. Dry running destroys it in minutes. There is a low-level interlock that should also hold it off, but it is a backstop, not a plan.
2. **Chlorinator off.** Fresh water has no salt, and electrolysis at low salt drives the cell voltage up and damages it. It will show *Check Salt* until the salt is back, which is correct rather than a fault.
3. **Reintegro on ①, all four ④ valves open.** The top-up discharges into the return manifold, so mains water reaches the pool through the four wall outlets. **It does not fill the tank.** On ① it stops itself when the tank floats satisfy; on ④ you have forced the solenoid open with no cut-out.
4. **Leave the Kripsol lever alone unless the gauge says otherwise.** Kripsol's manual says to fill from the mains on ⑥, closed, so the water bypasses the filter. In this installation that is not needed: with the lever on ① and the mains flowing, **the gauge reads zero**, so nothing is being over-pressurised. Check the gauge; if it shows pressure with the pump off, go to ⑥. Otherwise save the lever movement.
5. **Expect a day or more.** This line delivers roughly 1.5 to 2 m<sup>3</sup>/h, so 72 m<sup>3</sup> takes 36 to 48 hours. To measure your own rate: one centimetre of pool level is about 400 to 500 litres, so time the rise over an hour. If it is slower than it should be, the **manual bypass loop** past the solenoid is the biggest single improvement available, and the **strainer** on the top-up line is the most likely restriction after a drain. **Close the bypass before the pool crests**, because unlike the reintegro it will not stop itself.
6. **While it fills, hold the standing water.** It has no sanitiser and no circulation. Pre-dissolve **2 to 2.5 g per m<sup>3</sup> of water actually in the pool** of the granular chlorine, so about 150 g at 60 m<sup>3</sup>, and pour it round the perimeter. Repeat every two or three days. That costs about 1 mg/L of cyanuric acid, against 8 for a full shock. **Write it in LOG.txt**, with the amount.
7. **Watch the clear column, not the pool.** The instant it starts to rise, the pool is spilling and the tank is filling. From there it is **about 45 minutes to an hour to reach 0.55 m**, your start level, and roughly another hour before the reintegro satisfies and shuts off.
8. **Kripsol's pre-start checks.** Turn the motor's fan blades by hand from the back to confirm it is not seized after standing dry. Check the thermal overload has not tripped. Open the suction and discharge valves and **fill the pump body with water by hand** through the clear strainer lid, and check that lid's O-ring, which is what stops it sucking air. The valve Kripsol means is **SUCTION**, the red handle immediately before the strainer; no valve is visible on the discharge, so here there is one to open, not two.
9. **Valves:** ⑥ open, which is the same valve as ⑦, all four ④ open, ② and ③ open, ① almost closed, ⑤ and ⑧ shut, filter drain ⑨ shut, and the level column's **ISOLATOR** open so the column reads the tank rather than a level it was left at.
10. **First start on ⑤, circulation, not ①.** Kripsol's instruction, and it is not the obvious choice: circulation bypasses the filter, so you are not driving air and disturbed debris through the sand on the first run. Panel off before the lever moves, handle pressed down as a clutch. **Watch the strainer lid clear of air within a minute or two. If it is still churning, stop.**

11. Then ② backwash until the sight glass runs clear, two to three minutes, then ③ rinse briefly, then ① filtration. Panel off for every movement. **Write the gauge reading into LOG.txt after that last backwash:** it is the new clean baseline. On 2026-09-09 that was **1.18 bar**.
12. **Salt: about 290 kg, twelve 25 kg sacks**, added with the pump running so it dissolves and circulates. Twelve sacks lands 72.5 m<sup>3</sup> at 4.1 g/L. **Do not exceed 5.0 g/L**, the robot's ceiling. Only then switch the chlorinator on.
13. **Calibrate the pH probe** in pH 7.5 buffer. You have replaced the entire body of water, so whatever it was trimmed to no longer applies. Check the acid drum has something in it before the TRi PRO starts dosing.
14. **Keep the robot out** until free chlorine is under 4 mg/L, which rules it out during any commissioning shock.

**Nothing here can flood.** The pool's level is capped by the deck channel, the channel drains to the tank, and the tank has a passive high-level overflow to waste. A stuck float sends mains water quietly to the drain rather than onto a floor. *[added]* The one exception is the manual bypass, which has no cut-out at all.

## ADD SALT TO THE WATER

**The rule that governs everything here: salt cannot be taken out.** Nothing consumes it and nothing removes it except dilution, which means draining and refilling. So **under-add and top up**, exactly as with a flocculant. The window is narrow at both ends: below **4.0 g/L** the cell runs at rising voltage and damages itself, and above **5.0 g/L** the robot cleaner is out of specification.

1. Work out the charge. At **70 to 75 m<sup>3</sup>**, 4.0 g/L needs **280 to 300 kg**, and one 25 kg sack raises this pool by about **0.345 g/L**.
2. **Pump running, Kripsol on ①**, filtration, for the whole operation and for hours afterwards.
3. **Put it in the pool, never in the balance tank.** Undissolved salt in the tank is drawn straight into the pump suction and then the cell.
4. Shallow end, **spread out rather than piled**, and brush it along to help it go. Granular dissolves in minutes to hours.
5. **Tablets go in a mesh bag or a clean plastic basket**, weighted, in the shallow end. Nothing solid can then travel, nothing sits as a concentrated heap against the resin, and you can lift the bag to see what is left. Allow a day or two: slow dissolution is what the format is designed for.
6. **Add most of it, not all of it.** Hold back two or three sacks, let the water circulate for several hours, then read the salt figure on the TRi PRO and add the balance against a measurement rather than an estimate.
7. Only when the reading sits **between 4.0 and 5.0** does the chlorinator go on. Until then **Check Salt** is the correct display, not a fault.
8. **Top up after backwashes.** Each backwash and rinse loses 900 to 1,500 L and with it 3.6 to 6.0 kg of salt, so one sack replaces four to seven of them.

## EMPTY THE POOL

1. Pump off. Kripsol lever **M** to **④** (Waste). Valve **①** vertical (open).
2. Valve **⑦** horizontal (closed). If you leave it vertical (open) the balance tank empties as well.
3. Valve **L** open (pool floor suction). With **M** on **④** the pump draws off the bottom and sends it to waste, which is what empties the pool.
4. Pump control to **M** (manual).
5. Top-up control to **0**.

## §4Routine care

What running this pool actually asks of you, and how often. The printed sheets covered procedures but never the rhythm, which is the part nobody tells a first-time owner. Two things are worth understanding before the table: most of this takes seconds rather than minutes, and **the most consequential items are the ones with no symptom until it is too late** - a probe that has drifted, a drum that has run dry, an RCD nobody has ever tested.



| HOW OFTEN                 | WHAT                                                                                                                                                   | WHAT A FIRST-TIMER MISSES                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Walking past              | Glance at gauge ④, at the clear column, and at the TRi PRO display.                                                                                    | Read the gauge <b>against 1.18</b> , this installation's clean baseline, not against a number from a manual. A column that never moves is not a calm pool: it usually means the <b>ISOLATOR</b> at its foot is shut, and then the top-up has quietly stopped working with nothing else in the room looking wrong.                                                                                                                                           |
| Weekly                    | Test pH and free chlorine. Empty the pump strainer basket. Rinse the robot's cartridges. Brush the waterline. Check the acid drum has something in it. | <b>Rinse the cartridges after every run, not when they look full</b> - a loaded cartridge stops filtering and the robot just drives around. <b>Brush the waterline</b> : oils and film collect in that band and it is where scale and biofilm start, and on black resin you can see it. And an <b>empty acid drum is silent</b> - the peristaltic pump keeps running, the display keeps saying it is dosing, and the pH climbs with nothing to announce it. |
| Monthly                   | Read the salt figure on the TRi PRO. Look at the cell for scale. Look into the balance tank through the hatch, and at the overflow channel for leaves. | <b>Salt is never consumed</b> , only lost - through backwash, splash-out and draining. So a falling salt reading is not the cell using it up, it is water that left. Top up against the reading rather than a schedule.                                                                                                                                                                                                                                     |
| Every two months          | Calibrate the pH and ACL probes in buffer.                                                                                                             | The single most-skipped job on a salt pool. <b>Probes drift</b> , so the pH the TRi PRO believes it is holding may not be the pH in the water, and every automatic acid dose is then aimed at the wrong target.                                                                                                                                                                                                                                             |
| Every six months          | Press the test button on the 30 mA RCD.                                                                                                                | It is the life-safety device in a wet underground room, and nothing in the records says it has ever been tested.                                                                                                                                                                                                                                                                                                                                            |
| Each season start         | Run the sump pump deliberately. Open the column's ISOLATOR and watch the level move. Test cyanuric acid.                                               | The sump pump is the only way water leaves that floor, so the first you would know of a failure is standing water over the motor and the panel. Watching the column move is the <b>only</b> check that proves the level this room reads is the level in the tank.                                                                                                                                                                                           |
| Yearly, or on the reading | Descale the cell. Review the sand.                                                                                                                     | The bed was renewed about September 2025, so sand is years off. Cell scale is the one that creeps.                                                                                                                                                                                                                                                                                                                                                          |

## FOUR THINGS THAT HAPPEN ON A TRIGGER, NEVER ON A CALENDAR

- **Backwash** when gauge ④ reaches about **1.5**, not on a schedule. Washing a clean bed throws away 900 to 1,500 L, the salt in it, and any flocculant layer, and buys nothing.
- **Top up the salt** after backwashes, because that is when it leaves. One 25 kg sack replaces four to seven backwashes.
- **Re-dose flocculant** after each backwash while a clarifying treatment is running, because the backwash is what removes it.
- **Dose chlorine** on a test result, not on a feeling. While the cell is off this is the only thing keeping the water safe.

## HABITS THE SHEETS NEVER MENTIONED

1. **Calibrate the pH and ACL probes at least every two months during the season**, in pH 7.5 buffer. Probes drift, so the pH the unit believes it is holding may not be the pH in the pool.
2. **Press the test button on the 30 mA RCD every six months.** It is the life-safety device in a wet underground room and nothing in the records says it has ever been tested.
3. **Write down the gauge reading straight after every backwash.** That is this filter's clean baseline, and the climb away from it is the real signal that the sand is loading.
4. Move the valve as Kripsol says: motor stopped, press the handle down firmly first because it acts as a clutch, then move it.

**Note:** the level column's **ISOLATOR**, the red-handled valve at its foot, cuts the column off from the balance tank. Closed, the column freezes at whatever level it was showing and the automatic top-up quietly stops working, with nothing else in the room looking wrong. **Open it and watch the column move** at the start of each season, which is the only check that proves the level this room is reading is the level in the tank.

**§5Water and salt**

| TARGET                    | VALUE                                        | FROM THE SHEETS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pH                        | 7.2 to 7.4                                   | Below 7 is acidic, above 7.6 is basic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Free chlorine             | 0.6 to 1.5                                   | 1 is ideal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Robot's water limits      | Cl max 4 mg/L<br>pH 7.0 to 7.8<br>6 to 35 °C | From the Maytronics manual. Your normal targets sit inside all three. Two places they bite: <b>a commissioning shock at 10 to 15 g/m<sup>3</sup> gives about 8 mg/L, double the robot's chlorine limit, so take the robot out and leave it out until free chlorine is back under 4;</b> and a black resin pool in an Italian summer can approach the 35 °C ceiling. Below 15 °C its climbing suffers, which is the same threshold at which the sheets say to switch the electrolyser off.                                               |
| Effect of the dark finish | runs warm                                    | A black surface absorbs solar heat, so this pool sits warmer than a pale one. Three consequences worth knowing: <b>chlorine is consumed faster</b> , so the cell works harder; <b>evaporation is higher</b> , so the top-up runs more and carries away more salt; and <b>calcium is less soluble in warm water</b> , so it scales more readily, which is what the Svernante on the shelf is for. Scale also shows up starkly as white film on black resin, which is a useful early warning you would not get on a pale pool.            |
| Filter pressure           | about 1.18 bar                               | <b>Backwash on the climb away from 1.18, around 1.5</b> , which is where the dial turns from green to red. Record the reading straight after every backwash: that is the baseline, and the climb away from it is the real signal. <b>Headroom is tight:</b> the vessel is rated 1.6 kg/cm <sup>2</sup> , so there is only about 0.4 bar between a clean bed and the plate limit. Never let it approach 1.6. Measured on site 2026-09-09, superseding Kripsol's generic 0.5 to 0.7, correcting the sheet's “about 1 bar”.                |
| Salt                      | 4.0 to 5.0 g/L                               | <b>Narrower than the sheets say, and the reason is the robot.</b> The sheets give 4 to 9 g/L and Zodiac asks for above 4. But the Maytronics manual states a hard maximum of <b>5,000 ppm, that is 5.0 g/L</b> , above which the robot is out of specification. The window for running both machines is therefore 4.0 to 5.0. At 4 g/L a 72.5 m <sup>3</sup> pool needs about <b>290 kg, twelve 25 kg sacks</b> ; twelve sacks lands you at 4.1 g/L, comfortably inside both limits. Do not aim for the middle of the old 4 to 9 range. |
| Salt top-ups              | 25 kg sacks                                  | Two or three at a time when the unit signals. Only modest additions are needed after the initial charge: salt is lost to backwashing, not to evaporation.                                                                                                                                                                                                                                                                                                                                                                               |

| TARGET                          | VALUE      | FROM THE SHEETS                                                             |
|---------------------------------|------------|-----------------------------------------------------------------------------|
| pH down dosing                  | about 1 kg | Per 0.2 of pH reduction.                                                    |
| Electrolyser, water below 15 °C | off        | Pointless: algae and bacteria growth is reduced or stopped.                 |
| Electrolyser, water above 15 °C | on         | Needed: bacteria, fungi and algae reproduce quickly at higher temperatures. |

#### THE ONE RULE THE SHEET PUTS IN BOLD

- **Correct pH first.** Then the others, one at a time. Never adjust two things together.

The sheets also cover: clear water with algae, shock chlorinate and brush the floor; cloudy water with algae, the same plus check the gauge and backwash; cloudy water alone, test and consider flocculant, which clumps fine particles so the filter can catch them. Higher temperature and more swimmers both call for more chlorination, either the BOOSTER function or manual product.

## §6When something goes wrong

None of this was on the printed sheets. The fault table is from the makers' own manuals in *10\_manuals/*; the symptom guide below pairs each problem with a product you actually have, worked out for this pool. Full detail in *11\_products/CATALOGUE.txt*.

| WHAT THE TRI PRO SHOWS | WHAT IT MEANS                                                                                    | WHAT TO DO                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| No Flow                | Too little or no water through the cell.                                                         | Production stops on its own. Check the pump is running and the cell is not scaled. The Flow LED lit with the pump off is normal, not a fault. |
| Check Salt             | Salt between 3,000 and 4,000 ppm, depending on water temperature.                                | Bring it above 4 g/L. One 25 kg sack raises this pool by 0.345 g/L.                                                                           |
| Output Fault           | Possible power supply problem.                                                                   | Kill the power at the outlet and call someone.                                                                                                |
| Cleaning               | The cell is reversing polarity. Not a fault.                                                     | Wait about five minutes.                                                                                                                      |
| pH Low<br>pH High      | Empty drum, or a probe needing cleaning or calibration. High: a circulation or probe fault.      | Check the acid drum, the peristaltic tube, and when the probe was last calibrated.                                                            |
| pH ERROR               | Dosed a full set of cycles without reaching set point, <b>or has not run for over 72 hours</b> . | Note the second half. This one fires from inactivity too, not only from failure.                                                              |
| ACL Low<br>ACL High    | Under- or over-production, or a circulation problem.                                             | Check salt, cell condition and flow before anything else.                                                                                     |
| ACL ERROR              | Thirty cumulative hours producing without reaching set point, or thirty hours idle.              | Also fires from inactivity.                                                                                                                   |
| pH ---                 | pH regulation is not enabled yet. Normal.                                                        | It enables itself about eight hours after power-on.                                                                                           |

| SYMPTOM                  | CAUSE                                                                                                                                                                                                                                            | WHAT TO REACH FOR, FOR THIS POOL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clear but faintly cloudy | <p>Fine particles passing straight through the sand. This pool's weak point, because the bed runs at a high velocity.</p> <p><b>Not tired sand:</b> it was renewed about September 2025, so age is not the cause and a flocculant will hold.</p> | <p><b>Measure the cloudiness before treating it.</b> A black pool cannot be judged along the surface, because at a shallow angle you see reflected sky rather than water. It has to be looked at straight down against a submerged white target, and this pool owns two: the <b>white robot cleaner</b> on the floor and the <b>round white floor grate, about 20 cm across</b>. Read them on a fixed scale so today can be compared with tomorrow: <b>clear</b> means a crisp outline with the grate's slots visible; <b>slight haze</b>, a solid white shape with soft edges; <b>cloudy</b>, a pale patch with no defined edge; <b>very cloudy</b>, you know where it is but cannot see it; <b>opaque</b>, you cannot find it. Note the depth alongside the reading, and read it in flat light rather than bright sun, which puts glare on a dark surface. Running the robot also gives a live readout of the cause: if the grate sharpens as the cartridges load, the cloudiness was particulate; if hours of running leave it unchanged, suspect calcium, because no cartridge filters something still in solution. <b>Then check the gauge</b>, against the <b>1.18 bar baseline</b> rather than any absolute number. Near 1.5 the filter is the cause and no chemical will fix it; sitting at 1.18 it is not the filter. Otherwise flocculant: 1,088 g pre-diluted into the balance tank, backwash every 4 to 5 hours until clear, then 72.5 g each evening after every backwash. <b>Dose it only with the pump running, and leave the</b></p> |

| SYMPTOM | CAUSE | WHAT TO REACH FOR, FOR THIS POOL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |       | <p><b>pump running afterwards.</b> The balance tank is not a mixing vessel, it is a stop on the circuit: nothing moves through it unless the pump is drawing, because the pool spills into it over an open pipe with an air break and the floor suction (L) is shut. Dosed into a still tank, a flocculant coagulates that tank's own water instead of the pool's, and what it forms settles. The tank's outlet sits <b>0.35 m above its floor</b>, so anything reaching the bottom is below the pump's reach and stays there. The dose is not lost, but it is spent on the wrong 3 m<sup>3</sup> and a share of it is spent for good. <b>Start below the label dose and top up, rather than committing the full amount.</b> The asymmetry is what matters: too little coagulant is correctable by adding more, while too much reverses the particle charge, re-stabilises the haze, and can only be undone by filtering it back out. So dose low, watch the gauge climb and the halo tighten over 4 to 5 hours, and add the balance if neither moves. Note also that the 1,088 g figure is 15 g/m<sup>3</sup> times an <b>estimated</b> 72.5 m<sup>3</sup>: until the pool is measured the "full dose" is itself uncertain by a third either way.</p> <p><b>Two things to get right on timing.</b> Do not flocculate while anything is still growing or while rain is running in, because you would be clarifying water that is being re-dirtied; sanitise first. And <b>flocculate before the salt goes in if you have the choice</b>, because the treatment needs a backwash every 4 to 5 hours and each backwash costs 3.6 to 6.0 kg of</p> |



| SYMPTOM                                 | CAUSE                                                                                  | WHAT TO REACH FOR, FOR THIS POOL                                                                                                                                                                                                                                                       |
|-----------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |                                                                                        | salt. <b>Also run the robot with its ultra-fine cartridges.</b> It filters independently of the sand and catches the size of particle the bed is passing, which makes it a standing remedy rather than a dose. Rinse its cartridges after every cycle or it stops picking anything up. |
| Cloudy, hard water, visible scale       | Calcium coming out of solution.                                                        | Svernante, 725 to 1,450 g. Only once the water is already clear, or it is spent flocculating dirt instead.                                                                                                                                                                             |
| Green, or slippery walls                | Algae.                                                                                 | Algaecide, 1.45 L initial then 363 mL weekly. Expect foam, and expect the cell to work harder while a quat is in the water.                                                                                                                                                            |
| Chlorine low though the cell is running | Either the cell is not producing, or the chlorine is locked up by cyanuric acid.       | Work the fault table above first. Then measure cyanuric acid: above 50 mg/L is lock-up, and the target outdoors is 25 to 30. <b>The cure is dilution, not another chemical.</b>                                                                                                        |
| pH keeps climbing                       | Normal on a salt pool: electrolysis drives pH up. The acid drum should be handling it. | Check the drum, the peristaltic tube and the probe calibration before dosing by hand. Then pH minus, 290 to 435 g per 0.1 unit.                                                                                                                                                        |
| Pump will not start                     | On an overflow pool the pump                                                           | Check the tank level on the clear column first, then the motor thermal reset, the C16 MCB and the RCD. The outlet is at 0.35 m, so there is                                                                                                                                            |

| SYMPTOM    | CAUSE                                                                                                                                                                                                                             | WHAT TO REACH FOR, FOR THIS POOL                                                                                                                                                                                                                |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | draws from the tank, so a <b>low-level interlock</b> holds it off when the tank is too low. On  you bypass the time clock but not the interlock. | nothing to draw below that; start at about 0.55 m. <b>Do not defeat it:</b> if the column's <b>ISOLATOR</b> is shut the floats read satisfied and the interlock will permit a dry run. That valve is on the drawing, at the foot of the column. |
| pH too low | Usually over-correction.                                                                                                                                                                                                          | pH plus. <b>The two tubs are not interchangeable by volume:</b> 290 to 435 g of the Lapi against 1,450 g of the NortemBio, for the same 0.1 unit.                                                                                               |

## THE TWO TRAPS IN THE RECUPERO BOTTLE

- It is sodium bromide, and it exists to recover oxidising power once cyanuric acid passes 50 mg/L. A part-used bottle is evidence this pool has had chlorine lock at least once.
- Its own shock step calls for an unstabilised chlorine, calcium hypochlorite. **You do not have any.** Using the granular chlorine you do have would add more cyanuric acid and deepen the very problem you were treating.
- Its target pH is 7.5 to 7.7, above the 7.2 the TRi PRO holds, because bromine chemistry wants the higher figure.
- Sodium bromide in a salt pool turns the cell into a bromine generator, and the bromide stays until it is diluted out. Worth one technician's opinion before reaching for it again.

## §7The chemical stock

Nine products on the shelf, plus the acid drum in this room. Every one is photographed with its label in *11\_products/*, and doses below are the label's own rate applied to 72.5 m<sup>3</sup>. *CATALOGUE.txt* in that folder carries the lot numbers, the hazard statements and the working.

| PRODUCT                        | WHAT IT ACTUALLY IS                                                       | SIZE              | FOR THIS POOL                                                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| pH Più granulare Lapi          | <b>Sodium carbonate</b> , CAS 497-19-8. Soda ash.                         | 25 kg             | 290 to 435 g per 0.1 pH up. <b>Also the correct neutraliser for an acid spill</b> , applied gradually. Never caustic soda.                       |
| pH+ Increaser NortemBio        | A second, different pH raiser.                                            | 6.5 kg            | 1,450 g per 0.1 pH up. Roughly four times the Lapi rate for the same job.                                                                        |
| pH Meno granulare Lapi         | <b>Sodium bisulfate</b> , CAS 7681-38-1.                                  | 8 kg              | 290 to 435 g per 0.1 pH down. Not the gentle alternative to the acid drum: for the same movement it leaves about 2.15 times the sulphate.        |
| Cloro granulare 56% Acquaverde | <b>Sodium dichloroisocyanurate</b> , CAS 51580-86-0. Stabilised chlorine. | 5 kg              | Shock 725 to 1,088 g, pre-dissolved into the balance tank. Not for routine use: the cell makes the chlorine. Every dose also adds cyanuric acid. |
| Flocculante Klep               | <b>Polyaluminium chloride</b> , under 20%.                                | 5 kg              | Curative 1,088 g, preventive 72.5 g. The right tool for this pool's particular weakness.                                                         |
| Antialghe Axton                | <b>Benzalkonium chloride</b> at 4%, CAS 68424-85-1.                       | 5 L               | Initial 1.45 L, then 363 mL weekly. Foams, and consumes chlorine.                                                                                |
| Recupero Controlchemi          | <b>Sodium bromide</b> .                                                   | 5 L, a third full | Shock 725 g. Read the warning in the                                                                                                             |

| PRODUCT                                   | WHAT IT ACTUALLY IS                                                                                                           | SIZE        | FOR THIS POOL                                                                                                                                                                                                                             |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                                                                                                                               |             | previous section before using it at all.                                                                                                                                                                                                  |
| Svernante Controlchemi                    | <b>Aminotrimethylene phosphonic acid</b> , CAS 6419-19-8. A phosphonate scale inhibitor.                                      | 5 kg        | Winter 3.6 L, so about one container a season. Water must be clear first, pumps running 4 to 8 hours before you stop the plant.                                                                                                           |
| Sale Poseidon Atisale                     | Sea salt, no additives, <b>EN 16401 Quality A</b> , sold for electrolysis pools. The right salt, and those two marks are why. | 25 kg sacks | About 12 sacks for a full charge to 4 g/L, which lands at 4.1. One sack raises this pool 0.345 g/L. <b>Do not go above 5.0 g/L</b> : that is the robot cleaner's hard limit, and it caps the range well below the 9 g/L the sheets allow. |
| Sulphuric acid 36%<br><i>in this room</i> | The pH minus the TRi PRO doses automatically. Battery electrolyte grade, not a pool product.                                  | 25 kg drum  | About 1 kg per 0.2 of pH reduction. See the safety section.                                                                                                                                                                               |

#### FOUR THINGS THE SHELF IS MISSING

- CYA test** A cyanuric acid test. Three products on that shelf depend on knowing this number and none of them can tell you it. Without it the Recupero is guesswork.
- shock** An unstabilised chlorine, calcium hypochlorite or similar. Both the Recupero and the Svernante labels call for one and there is none here.
- sulphate** A sulphate test, given years of sulphuric acid on a resin-finished pool with no exposed grout, so this is a low-priority reading rather than the risk it first looked like: it is protecting the balance tank's coating and any place the pool's resin has failed.
- soda ash** Some of the 25 kg of soda ash you already own, kept **in this room** for acid spills. It is simply on the wrong side of the building.

## §8Safety notes

### CORRECT THIS ON THE PRINTED SHEET

- The fifth note on the operating sheet says "*Il bidone del cloro*", the chlorine drum. It is not chlorine. The container is **sulfuric acid, ACIDO SOLFORICO 36%**, confirmed from the label. On a salt pool the chlorine is made by the cell; the jug is the pH minus that the TRi PRO doses.
- **Never let anyone add a chlorine product to that jug, or store one beside it.** This is not general caution. Your granular chlorine carries **EUH031 on its own label: contact with acids liberates a toxic gas**. The gas is chlorine. The room is 4.52 m<sup>2</sup> holding roughly 10 to 11 m<sup>3</sup> of air, below ground, with one door. The wording on the sheet is exactly the kind of thing that could cause that mistake, so it is worth correcting the print-out by hand.
- **The white bucket in photograph 34 of 04 *plant-room*/ is marked UN 3077**, which is the granular chlorine's transport code. Whatever brought it into this room, it should not stand here. Keep the chlorine and the acid in different rooms.
- Handling: H314, causes severe skin burns and eye damage. Gloves and eye protection when changing the drum, and never add water to acid. Keep **sodium bicarbonate or soda ash** down here for spills, applied gradually, and **never caustic soda**, which reacts violently with sulfuric acid. You already own 25 kg of soda ash: it is the pH plus on the shelf.
- **36% is stronger than pool practice recommends.** PWTAG guidance is that sulphuric acid should never be supplied to pools above 25%, and that acids should generally be bought, stored and used below 15%. Separately, sulphuric acid above **15% w/w** is a restricted explosives precursor under EU Regulation 2019/1148, Annex I. That is a reason to ask your supplier, not a verdict from me. The practical answer is to move to a pool-grade pH minus at 15% or below when this drum runs low, and have the drum taken away properly rather than diluted down a drain.
- **Sulphate accumulates, but this pool is less exposed than it first appeared.** Sulphuric acid leaves sulphate behind, and the recognised limit of around **360 mg/L** exists to stop it attacking cementitious grout. **This pool has no grout:** it is finished in smooth black resin, so the mechanism that lifts tiles does not apply. What remains is the balance tank, whose coating is the only thing between the water and its concrete, and anywhere the pool's resin has failed and exposed the shell. Still worth one reading because nobody has ever taken one, but it drops a long way down the list. An earlier version of this document treated this as a tiled pool and rated it higher; that was my inference and it was wrong. The granular pH minus is still worse on this count than the acid, for the same pH movement.

## BECAUSE THE ROOM IS BELOW GROUND

- **The sump pump is the only drain.** Nothing leaves that floor by gravity. If it has failed, the first you will know is standing water over the pump motor and the panel. Worth running it deliberately at the start of each season rather than finding out.
- **Ventilation matters more here.** A 25 kg drum of sulfuric acid in a confined space below ground is a different proposition to the same drum in an airy outbuilding. Check the room has a working vent, and open the door for a while before working in there. **Nothing in this archive shows one:** the only opening in any photograph is a louvred grille that may be in the door leaf, which would vent to somewhere else inside the building rather than to outside air. It is in the open items, and it is the one there that could hurt somebody.
- **Salt electrolysis makes hydrogen** as a by-product. Normally it is carried away dissolved in the water, which is one more reason never to run the cell without flow. Isolate it on (B) and (C) whenever the pump is off for any length of time.

## IF WATER IS COMING IN: WHAT TO SHUT, IN ORDER

- The pool sits at ground level and this room is underneath it. That means the water upstairs is permanently trying to get down here, and a burst or an open joint will keep flowing long after the pump stops. **Switching the pump off does not stop it.**
- Shut, in this order: (7) (balance tank outlet), then (L) (pool floor discharge), then all four (D) returns. That cuts every path between the pool and this room. (G) as well if you can reach it.
- Then the sump pump (N) has a chance of keeping up. It is the only drain the room has.
- For a sense of scale: **the balance tank holds 3.69 m<sup>3</sup> in normal operation, which over this floor stands 0.82 m deep**, and 1.21 m if its safety overflow were blocked and it had filled to the coating line. The pool overhead holds roughly six times this room's entire air volume.
- This sequence is mine and is not on the sheets: it follows from the levels rather than from anything written down. It is worth checking with a technician, and worth knowing before you need it.

**§9Equipment**



| ITEM               | DETAIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SOURCE                         |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Pump               | DAB EuroPro 200 M, no. 2.1213, TF 60 S1. Q 31-29-26 m³/h at H 3-10-12 m, Hmin 5 m, max 2.5 bar. P2 2 HP, P1 1.9 kW. 220-240 V, 8.6 A, 2800 rpm, 50 Hz, class F, IP 55, 40 µF 450 V. App. no. NSW25167.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Nameplate                      |
| Filter valve       | Kripsol side-mount 6-way, 7 handle positions ( <b>0</b> and <b>1</b> to <b>6</b> ), listed in the procedures section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Valve body and sheets          |
| Sand filter        | <b>Sand renewed about September 2025</b> , so roughly one year into a five to seven year life. Recorded as new sand rather than washed sand; those are different clocks. Kripsol Ø760, Serie 12. Filtration area 0.45 m², rated flow <b>22 m³/h</b> , total sand charge <b>225 kg</b> , sand grading <b>0.4 to 0.8 mm</b> . Working pressure 0.5 to 1.5 kg/cm², test pressure 3.5 kg/cm², design filtration rate 50 m³/h per m². Kripsol, Ugena (Toledo), Spain.                                                                                                                                                                                                                                                                                                                                                                                                                 | Vessel plate                   |
| Flow, worth a look | The filter is rated 22 m³/h at 50 m³/h per m² of sand. The pump is rated 31, 29 and 26 m³/h at 3, 10 and 12 m of head. Read on their own those figures say the pump over-flows the filter, but the pump nameplate is the wrong number to use: what actually flows is set by the narrowest part of the circuit, and the tank draws through a single modest-bore line. At a sane suction velocity of about 2 m/s a 63 mm line carries roughly 18 m³/h and a 75 mm line roughly 26. <b>So on 63 mm you are under the filter's rating and the pump is simply oversized, which costs electricity and bearing life rather than damaging the filter. On 75 mm you are at the plate limit.</b> Measuring the suction pipe's outside diameter settles which. For reference, the DIN 19643 figure for single-layer sand is 30 m³/h per m², well below what this filter's own plate allows. | Plates, plus manufacturer data |
| Chlorinator        | Zodiac TRi PRO, salt electrolysis with pH and ORP regulation and an integral peristaltic dosing pump. Cell in a clear housing on a three-valve arrangement ( <b>B</b> , <b>C</b> , <b>A</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Front panel                    |
| Panel              | Tele-Matic enclosure. <b>Upper row, the pump:</b> Hager <b>MN513N</b> motor protector set 6.3 to 10 A, <b>ESC 225</b> contactor, the <b>M 0</b> <b>A</b> selector, <b>SVN 126</b> indicator, and the <b>EH011</b> 24 h time switch labelled <b>PROG. POMPA</b> . Then an <b>ST314</b> 40 VA 230 to 12/24 V auxiliary transformer, <b>LS501</b> 10.3×38 fuse holders, two <b>Lovato LVM 20</b> level relays for conductive liquids labelled                                                                                                                                                                                                                                                                                                                                                                                                                                       | Photograph 35                  |

| ITEM          | DETAIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SOURCE                                                           |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|               | <p>(<b>CONTROLLO MINIMO LIVELLO</b>) and</p> <p>(<b>CONTROLLO LIVELLO REINTEGRO</b>), and the reintegro's own selector. <b>Lower row: CDC 724H</b> 25 A 30 mA RCD, <b>MJN516A</b> C16 MCB, the lights, and a second round 24 h rider dial that no legend claims and whose function is not established.</p>                                                                                                                                                                                                                                                                                                                                 |                                                                  |
| Top-up        | <p>Mains through tap (<b>E</b>), a strainer, and solenoid (<b>F</b>) driven by the level relays. A manual bypass loop with a third blue valve lets you fill by hand if the solenoid fails. Discharges into the <b>return manifold</b>, not the tank. Whether the line carries a non-return valve is unverified, and it should: the pool sits above this room, so pool water could siphon toward the supply if mains pressure dropped.</p>                                                                                                                                                                                                  | Observed, plus the discharge point reported on site              |
| POD           | <p>Zodiac TRi PRO probe holder, clamped to the low horizontal pipe between the filter and the cell branch. One body carrying four things: the flow switch that stops the cell when there is no flow, the pH probe, the ORP probe, and the pH minus injection point. The manual requires it on a horizontal run so the probes hang vertically, at least 30 cm of straight pipe, more than 20 cm from an elbow, and <b>first after the filter</b>. This one meets all of that.</p>                                                                                                                                                           | Photograph 04 and 29, Zodiac TRi PRO manual section 3.3          |
| pH correction | <p>ACIDO SOLFORICO 36% (sulfuric acid), 25 kg drum, Siac srl, Torino. UN 2796, ADR class 8 packing group II, index 016-020-00-8, H314. Composition line reads "elettrolito per batterie ... acido solforico 30/31 Bé", so it is battery-electrolyte grade rather than a pool-branded product. Dosed automatically by the TRi PRO peristaltic pump, which draws from the drum and injects into the POD, before the cell. Sheet figure: about 1 kg per 0.2 of pH reduction.</p>                                                                                                                                                              | Drum label, photograph 04, Zodiac manual                         |
| Robot cleaner | <p><b>Maytronics Dolphin Maximus X70</b>, fitted with <b>soft Wonder foam brushes</b>, changed from the PVC ones in September 2026. That is the correct pairing: Maytronics specify foam for smooth surfaces and PVC for rough pebble or quartz render, and this pool is smooth resin. Cleans floor, walls and waterline on a 2.5 hour cycle. 17 m cable, rated for in-ground pools up to 15 m, so this 13 m pool is inside its range but not by much. 11.17 kg, supplied with a caddy, an ultra-fine filter kit and Wi-Fi scheduling through the MyDolphin Plus app. Four year warranty, so keep the proof of purchase. Water limits:</p> | Maytronics manual 8180035, in <a href="#"><i>10_manuals/</i></a> |

| ITEM     | DETAIL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SOURCE                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|          | <p><b>chlorine max 4 mg/L, pH 7.0 to 7.8, 6 to 35 °C, and salt max 5.0 g/L.</b> Power supply is switch-mode IP54, 100-250 V, 180 W, and must sit <b>at least 3.5 m from the pool edge and 11 cm off the ground</b>, roughly mid-way along the long side, with no extension cord. Its manual is badged <i>CLASSIC 8+ / TOP 8</i> and never says Maximus X70; it is nonetheless the right document. <b>It is completely independent of the plant room:</b> its own pump, its own filter cartridges, no connection to the circuit, which means it does not use the manual vacuum (H) or the vacuuming procedure in the procedures section at all.</p> |                       |
| Sump     | <p>(N), a floor gully with a submersible pump. The room is below ground, so the floor cannot drain by gravity and this pump is the only way water gets out. That makes it safety equipment rather than a convenience.</p>                                                                                                                                                                                                                                                                                                                                                                                                                          | Sheets, gully visible |
| Turnover | <p>At the filter's rated 22 m³/h, a 70 to 75 m³ pool turns over in roughly 3.2 to 3.4 hours. Overflow pools are normally run faster than skimmer pools, so that is a sensible figure. Actual flow depends on the real head and the state of the sand.</p>                                                                                                                                                                                                                                                                                                                                                                                          | Arithmetic            |

# §10What is still open

Nothing in this list is known and simply unrecorded. It is the honest inventory of what this document does not know, kept here so it can be closed off later rather than quietly forgotten. The third column of every row says what closing it would change, so you can see at a glance which are worth a trip to the plant room and which are only tidiness.

| PROCEDURES THIS DOCUMENT STILL OWES YOU |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |  | <p>Audited 2026-09-09 against everything that had been explained in conversation but never written down. Since written up: the backwash and what the filter is doing, cleaning the pump strainer, and adding salt. <b>Still only a sentence or a dose rather than a procedure:</b> winterising, four to eight weeks away and the reason the Svernante is on the shelf; calibrating the pH probe, which the TRi PRO manual covers and which falls due once the cell runs; changing the acid drum, 36% sulfuric and deserving steps rather than a caution; cleaning the robot's cartridges. <b>Absent altogether:</b> descaling the salt cell, which is core maintenance on a salt pool; opening the pool in spring; and changing the filter sand, years off since the bed was renewed about September 2025. None is urgent today, and all are the same defect the backwash had: a task you can follow but not reason about.</p> |

| MEASURE THIS                    | WHERE                                                                                                                                                                                 | WHAT IT CHANGES                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pool dimensions and volume      | The pool itself, with a tape, in daylight: length, width, and depth at both ends. The 10-minute column timing that would have given the surface area was not taken while it filled.   | <b>The salt charge, which is the next thing this pool needs.</b> 290 kg assumes 72.5 m³. At 80 m³ twelve sacks land about 3.75 g/L, under Zodiac's 4.0 minimum; at 65 m³ the same twelve land 4.6 and eat into the robot's 5.0 ceiling. Every dose in this document quoted per cubic metre moves with it. <b>The volume has never been measured. 70 to 75 m³ is an estimate.</b>                                                        |
| The pump clock, against a watch | The Hager <b>EH011</b> in the upper row, marked <b>PROG. POMPA</b> . Read its index against the real time, then read which of its riders are set and write both into <b>LOG.txt</b> . | <b>What <b>A</b> actually does.</b> On 2026-09-10 it read about 22:00 in a photograph timed 01:47, roughly four hours slow, so the filtration hours are not the hours anyone chose. It is a synchronous motor clock: it stops whenever the panel is switched off and never catches up, and the panel goes off for every Kripsol lever move, so it drifts again after every backwash. Nothing about it looks wrong from across the room. |
| What the second dial switches   | The round 24 h rider dial in the lower row, photograph 36, which no legend strip claims. Its riders are pushed down from 2 to 21.                                                     | Whether a timed circuit in this room is unaccounted for. It is <b>not</b> the filtration pump: that runs off the EH011 above it, which carries its own riders. The likely candidates are the lights and the chlorinator. To confirm, slide the EH011's own override to <b>I</b> and then <b>0</b> with the pump selector on <b>A</b> and watch whether the pump follows.                                                                |
| Suction pipe outside diameter   | The low pipe running from the compensation valve to the pump. 50, 63, 75 or 90 mm.                                                                                                    | Whether the filter is being over-flowed at all. On 63 mm it is not, and the pump is merely oversized, which costs electricity rather than sand. On 75 mm it sits at the plate limit. The equipment section currently states both cases because this is unmeasured.                                                                                                                                                                      |

| MEASURE THIS               | WHERE                                                                                                                                                                                                          | WHAT IT CHANGES                                                                                                                                                                                                                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ceiling height             | This room, floor to ceiling. The 3D scan puts a floor under it: its vertex density stops at 2.33 m, so the ceiling is <b>at least</b> that. It found no ceiling plane, because the phone was never pointed up. | The room's air volume, and with it whether the acid-storage warning in the safety notes is a measured fact or my judgement. Now known to be <b>10.5 m³ or more</b> . A tape would close it.                                                                                                                        |
| Column mark spacing        | The clear level column, distance between the marks 1 to 4.                                                                                                                                                     | Precision only. Multiply by 27.3 and the column reads in litres instead of positions.                                                                                                                                                                                                                              |
| Wall thickness             | Between this room and the tank.                                                                                                                                                                                | Precision only, and probably nothing: the 7 cm on the scaled plan is almost certainly a drawn separation. Worth confirming, because 5.46 m³ of water sits on the other side.                                                                                                                                       |
| What the resin actually is | The product name or datasheet for the black resin coating, from whoever applied it.                                                                                                                            | Its chlorine, pH and temperature tolerances. It matters before the commissioning shock dose, which will put roughly 8 mg/L of free chlorine into the water: most coatings take that comfortably, but you would want to know rather than assume. It is also what you would need if a patch ever has to be repaired. |

| LOOK AT THIS                            | WHERE                                                                                                                                                           | WHAT IT CHANGES                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What the ventilation actually is        | The louvred grille in photograph 26 of <a href="#">04_plant-room/</a> . Is it in the door leaf or in the wall beside it, and is there any other opening at all? | <b>Whether a 25 kg drum of 36% sulfuric acid sits in a room that vents to outside air, or to another space inside the building.</b> No wall vent, duct, fan or high-level opening appears in any photograph in this archive, and the one dark slot on the manifold wall turns out to be a cable chase. The safety notes say to check the room has a working vent. This is that check, and nothing here can answer it. |
| The 1.66 m left wall                    | Stand with your back to the door and photograph it.                                                                                                             | The plan draws nothing on it, because no photograph in this archive looks squarely at it: the open door leaf stands across it in the only image that can be read angularly. Blank on that drawing means unphotographed, not empty.                                                                                                                                                                                    |
| Whether the pump has a discharge valve  | The riser from the pump's volute up to the Kripsol six-way.                                                                                                     | Kripsol's pre-start check says to open <i>the suction and discharge valves</i> . Only <b>SUCTION</b> is visible in any photograph. If there is no valve on the discharge, the pump cannot be shut off on both sides for service, and that step means one valve rather than two.                                                                                                                                       |
| The multiport's open socket             | The white bore facing out of the Kripsol six-way, in photographs 16 and 25.                                                                                     | It reads as an open, unconnected port. If that is the waste port and it really is open, backwash water is not going where this drawing says it goes. Look behind and under the filter for what it connects to.                                                                                                                                                                                                        |
| The second penetration at the wall foot | The base of the manifold wall, immediately right of the hole the orange 110 leaves through. Photograph 24 shows a silver-taped stub in a mortared hole.         | The balance tank is behind the other wall, so this is not one of its four connections. Nothing in this archive says where it goes.                                                                                                                                                                                                                                                                                    |

| TEST THIS                                | WHY                                                                                                                                                                                                                                                                                                                | THRESHOLD                                                                                                                                                                                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyanuric acid                            | <p>The most consequential unknown in the whole chemistry. Three products on the shelf depend on this number and none of them can measure it. A part-used bottle of sodium bromide is evidence this pool has hit chlorine lock at least once, and above the threshold the cell's output is simply being wasted.</p> | above 50 mg/L is lock-up target 25 to 30 outdoors                                                                                                                                                                                                              |
| Sulphate                                 | <p>Years of sulphuric acid dosing, never measured. Lower priority than it looked: the pool is resin-finished with no grout, so the limit is only protecting the balance tank's coating and any place the pool's resin has failed. Cheap enough to do once.</p>                                                     | 360 mg/L                                                                                                                                                                                                                                                       |
| The first actual pH and chlorine reading | <p><b>Answered 2026-09-10: the kit exists.</b> A Lovibond tablet comparator, photographs 20 to 22, reading pH 6.8</p>                                                                                                                                                                                              | <p>Everything. Every target in the water section was unusable without it, and the holding doses through the refill all went in blind. <b>One caution before trusting a tenth of a pH unit:</b> the phenol red foil reads <i>PHOTOMETER</i>, which Lovibond</p> |



| TEST THIS                      | WHY                                                                                                                       | THRESHOLD                                                                                                                                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | to 8.2 and free chlorine 0.1 to 3.0 mg/L. The procedure is in the procedures section. What is still missing is a reading. | sells as a different article from the comparator tablet. Whether it develops the same colour on a visual card is not established here, and the kit's own leaflet will say.                                                   |
| Non-return valve on the top-up | Whether the mains feed into the return manifold has one.                                                                  | A drinking-water question rather than a pool one. The pool sits above this room, so if mains pressure ever dropped, pool water could siphon back toward the supply. I cannot tell from the photographs whether there is one. |

#### FIVE SMALL THINGS, NONE OF THEM EXPENSIVE

- 1. Move some soda ash into this room.** You own 25 kg of it as the pH plus on the shelf. It is the correct neutraliser for an acid spill and it is currently on the wrong side of the building.
- 2. Keep the chlorine out of this room, permanently.** Its own label carries EUH031: contact with acids liberates a toxic gas.
- 3. Clip the overflow pipe's open end** in the balance tank. It lies loose on the floor and will move when the pool spills hard.
- 4. Check the intercapedine is still ventilated.** That corridor is a cavity around the below-ground structure, so it is damp by design and the staining and blown plaster below the hatch are expected rather than alarming. It only stays harmless while the cavity can breathe.
- 5. Buy an unstabilised chlorine** for shock dosing, calcium hypochlorite or similar. Both the Recupero and the Svernante labels call for one and there is none on the shelf.

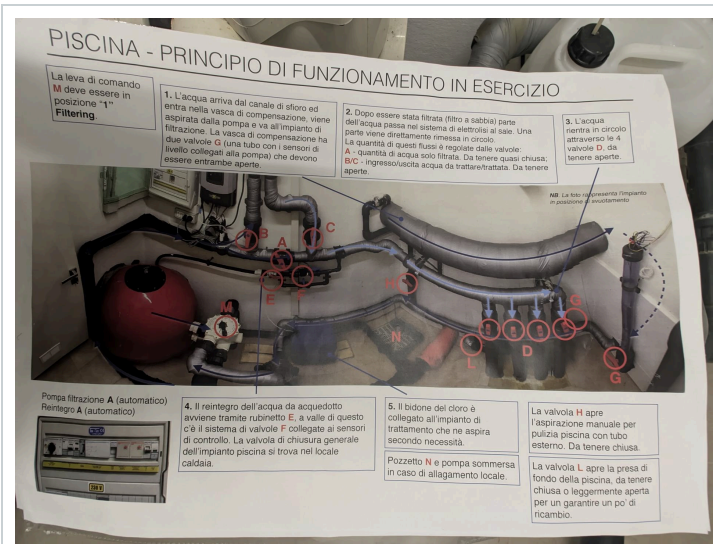
**Note:** everything above that needs the balance tank wants doing while it is empty and open. That will not stay true.

#### ONE THING NOBODY CAN MEASURE, ONLY FIND OUT

**dilution** **Has the pool ever been partly drained and refilled?** Backwash losses aside, nothing in the record says so. It bears directly on both figures in the table above: cyanuric acid and sulphate both accumulate and both are only removed by dilution.

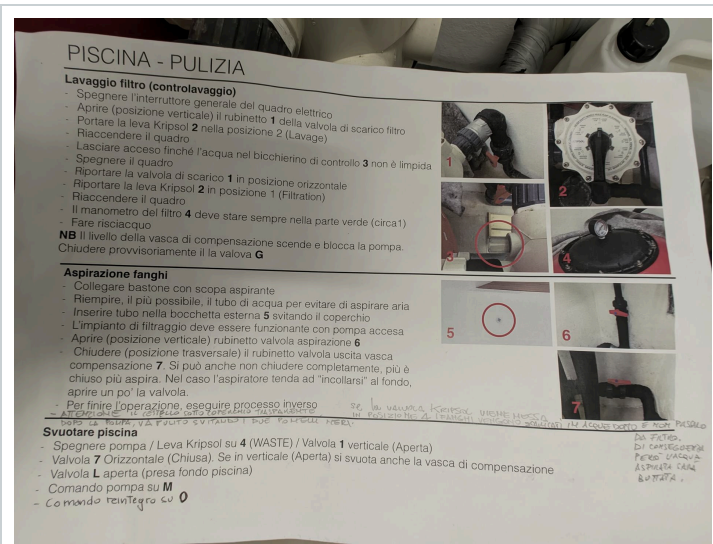
§11Printed documentation

The four printed pages that live in the room. Everything in the procedures, the water figures and the safety notes above is transcribed and translated from these. Where a maker's manual contradicts them, that is said at the point it matters. The filter gauge was the one case where this document followed the manual instead of the sheet, and on 2026-09-09 a reading taken on clean, freshly backwashed sand proved the sheet right: 1.18 bar, not Kripsol's 0.5 to 0.7. Kept here at full readability so the original is never more than one file away.



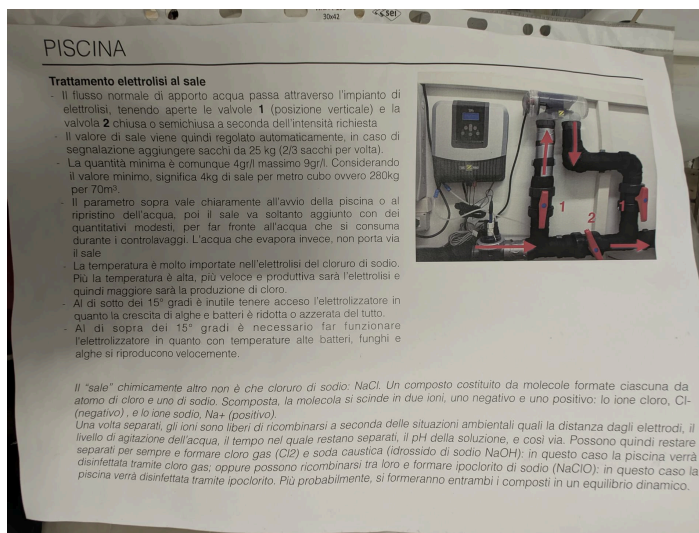
PRINCIPIO DI FUNZIONAMENTO

How the system runs. The source of the letters **(A)**, **(B)**, **(C)**, **(D)**, **(E)**, **(F)**, **(G)**, **(H)**, **(L)**, **(M)** and the sump **(N)**.



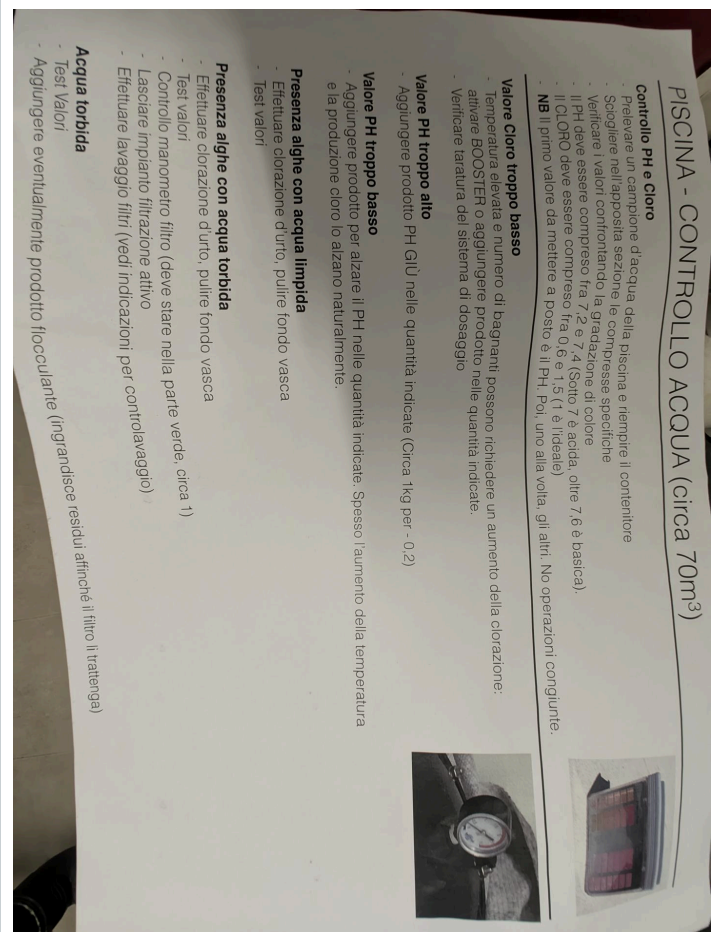
PULIZIA

Backwash, vacuuming and emptying, with handwritten notes in the margins.



## TRATTAMENTO ELETTROLISI AL SALE

Salt levels, temperature thresholds, and how the electrolysis works.



## CONTROLLO ACQUA

pH and chlorine targets and what to do when they drift.

## §12Reference photographs

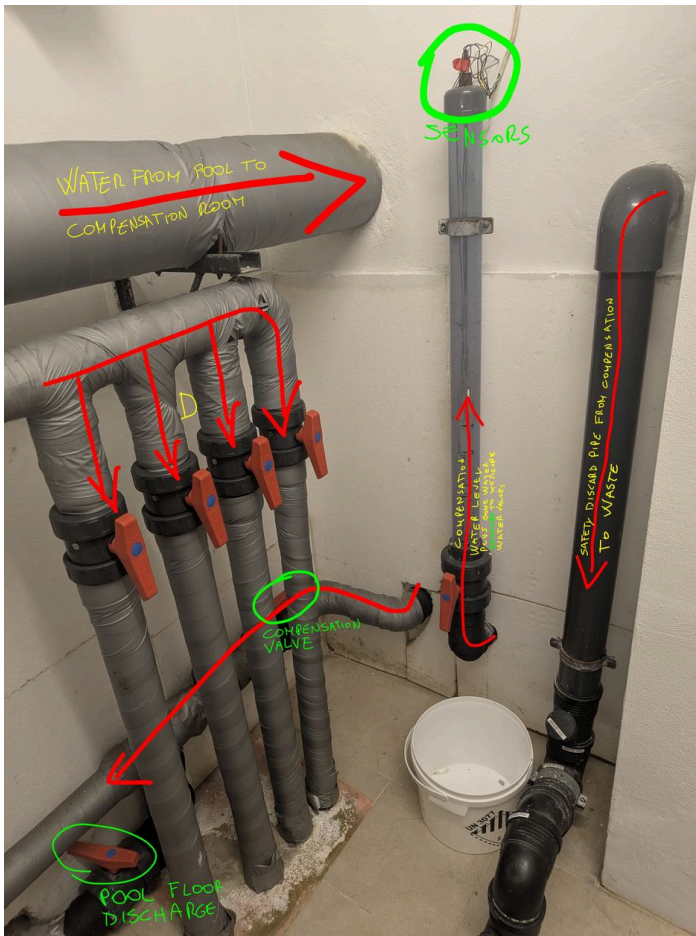
The images every factual claim in this document rests on. If anything here ever looks wrong, these are what to check it against.



### THE ROOM, 216 DEGREES

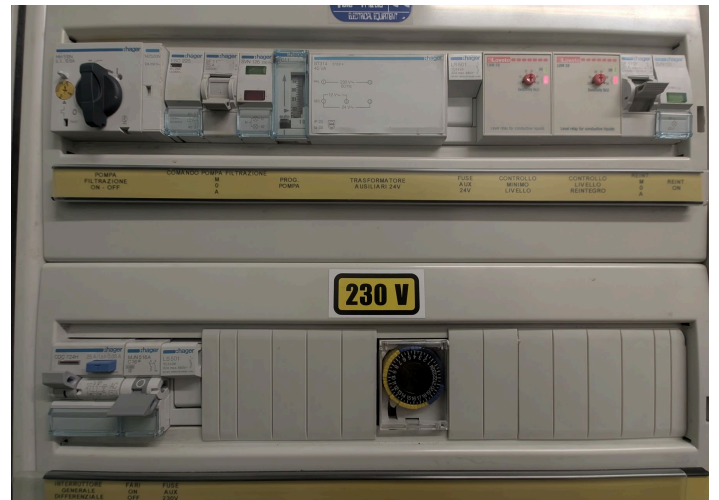
Phone photosphere, 39 megapixels. The best single record of the room as it stands.





### THE MANIFOLD WALL, YOUR NOTES

Your own annotations: the pool overflow heading to the compensation room, the sensors, the compensation valve, the safety discharge, and the pool floor discharge.



### THE ELECTRICAL PANEL

Photograph 35, which replaced 20 here on 2026-09-10 for being sharper: the Hager gear, the EH011 pump clock under **PROG. POMPA**, and the two LOVATO LVM 20 level relays.

| FILTRO ARENA                                                                                                                                         | SAND FILTERS                                                                                                                |       |       |       |       |       | FILTRE A SABLE |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|----------------|
| MODELO<br>MODEL<br>MODELE                                                                                                                            | Ø 40Q                                                                                                                       | Ø 450 | Ø 520 | Ø 640 | Ø 760 | Ø 900 |                |
| SUPERFICIE DE FILTRACION M <sup>2</sup><br>FILTRATION AREA M <sup>2</sup><br>SURFACE FILTRATION M <sup>2</sup>                                       | 0.13                                                                                                                        | 0.16  | 0.20  | 0.30  | 0.45  | 0.60  |                |
| CAUDAL M <sup>3</sup> /h<br>FLOW M <sup>3</sup> /h<br>DEBIT M <sup>3</sup> /h                                                                        | 6                                                                                                                           | 8     | 10    | 15    | 22    | 30    |                |
| CARGA ARENA Kg<br>TOTAL SAND Kg<br>CHARGE SABLE Kg                                                                                                   | 50                                                                                                                          | 70    | 100   | 150   | 225   | 325   |                |
| PRESION TRABAJO<br>WORKING PRESS<br>PRESSION TRAVAIL                                                                                                 | 0.5 - 1.5 Kg/cm <sup>2</sup>                                                                                                |       |       |       |       |       |                |
| PRESION PRUEBA<br>TESTING PRESS<br>PRESSION PRELIEVE                                                                                                 | 3.5 Kg/cm <sup>2</sup>                                                                                                      |       |       |       |       |       |                |
| GRANULOMETRIA ARENA<br>SAND GRADING<br>GRANULATIONS SABLE                                                                                            | 0.4 - 0.8 mm                                                                                                                |       |       |       |       |       |                |
| VELOCIDAD DE FILTRACION<br>FILTRATION RATE<br>VITESSE FILTRATION                                                                                     | 50 m <sup>3</sup> /h / m <sup>2</sup>                                                                                       |       |       |       |       |       |                |
| 1 - Aspiracion agua piscina<br>2 - 3 - Conexion bomba valvula<br>4 - 5 - Circuito filtracion<br>6 - Agua retorno a piscina<br>7 - Conexion a desaguo | 1 - Suction from pool<br>2 - 3 - Pump and valve connection<br>4 - 5 - Filtration<br>6 - Return to pool<br>7 - Waste connect |       |       |       |       |       |                |

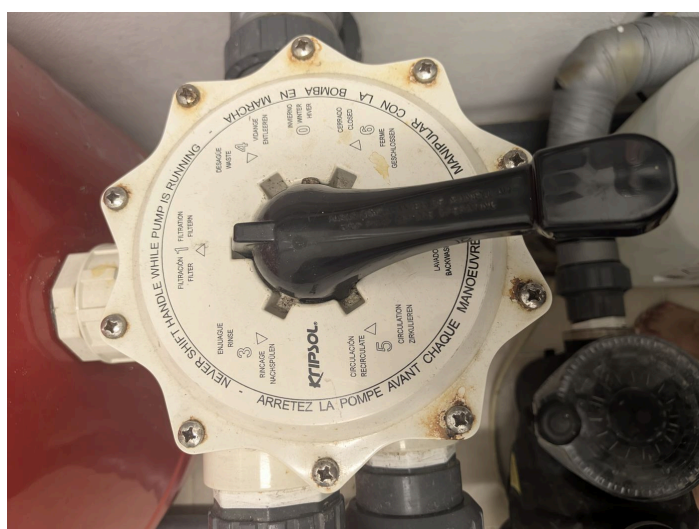
### FILTER DATA PLATE

Kripsol. The 760 column is the highlighted one: 0.45 square metres, 22 cubic metres an hour, 225 kg of sand, 0.4 to 0.8 mm.



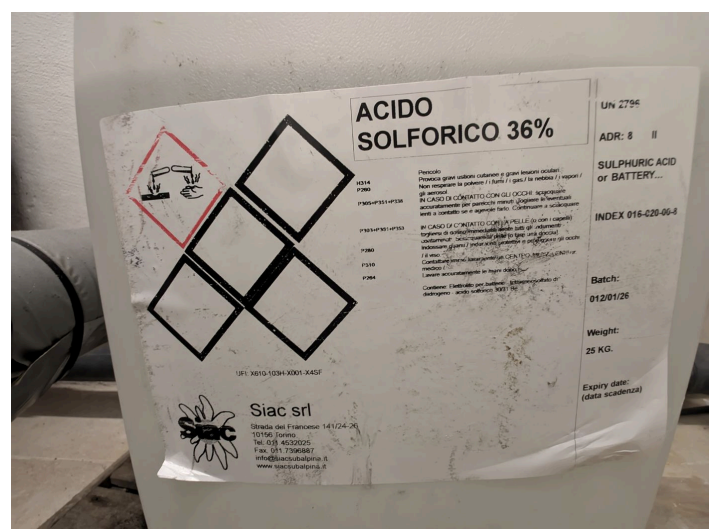
### PUMP NAMEPLATE

DAB EuroPro 200 M. 2 HP, 1.9 kW, 31 to 26 cubic metres an hour, 2800 rpm, IP55.



### KRIPSOL VALVE DIAL

All seven positions, and the warning cast into the rim in four languages.



### THE DRUM LABEL

ACIDO SOLFORICO 36 percent. UN 2796, class 8, H314. Sold as battery electrolyte.

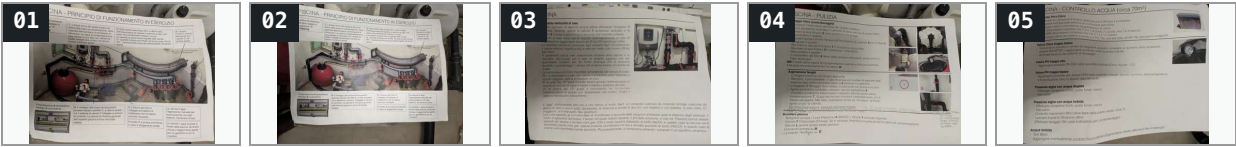


§13Every photograph, as an index

The whole library at thumbnail size, so this document can answer "which picture was that?" on its own, with no network and no repository to hand. Tap any one to enlarge it and read what it shows. These are reductions: the full-size original of every one is in the folder named above it, and the one-line description under each is the same text as its row in *MEDIA.tsv*, which is where those descriptions live.

02\_PRINTED-SHEETS/ 5 photographs

The four printed pages kept in the room. Everything in the procedures, the water figures and the safety notes is transcribed from these.



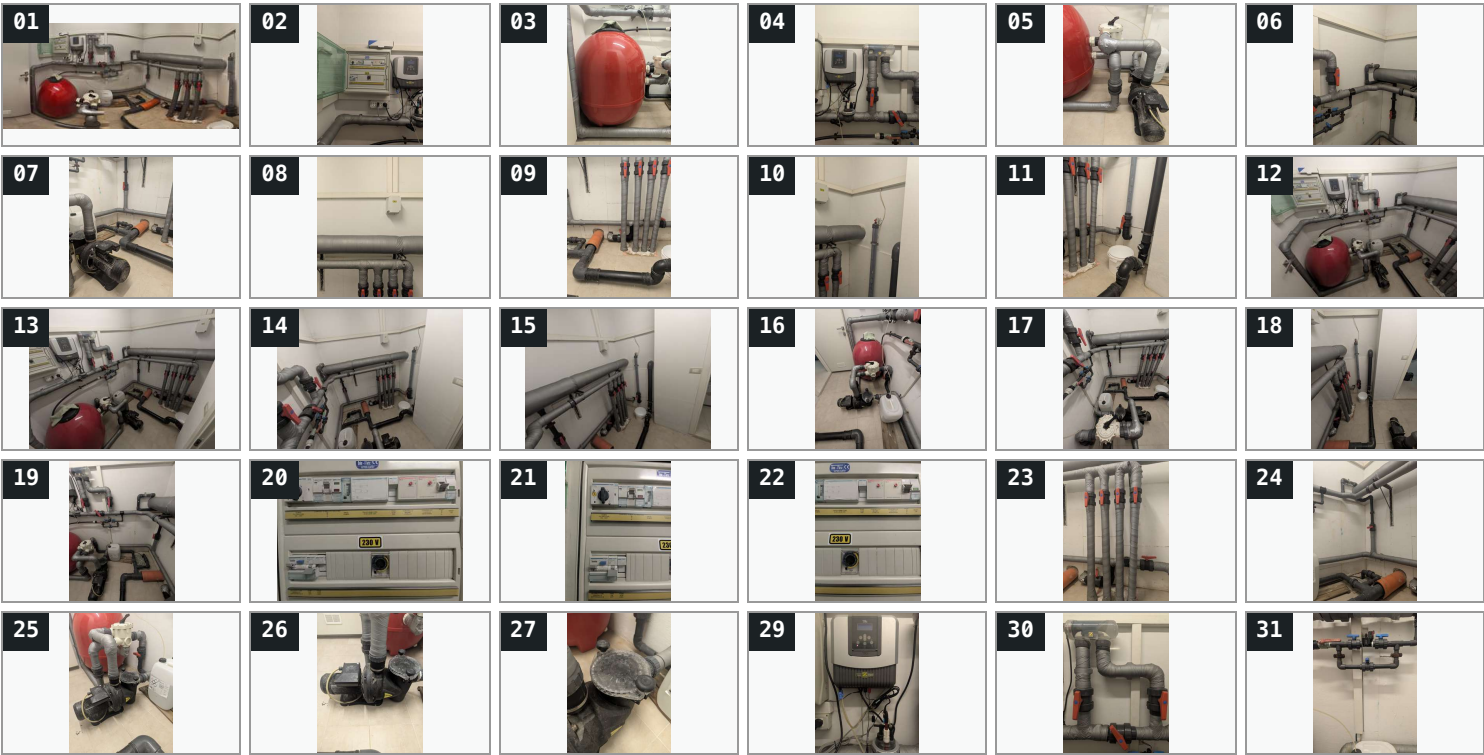
03\_NAMEPLATES/ 5 photographs

The plates and labels every specification in this document rests on. If a figure is ever in doubt, it is settled here.

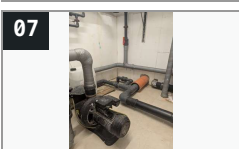


04\_PLANT-ROOM/ 36 photographs

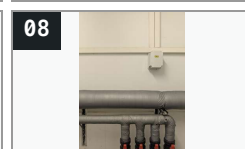
The evidence, in the order it was taken. Later numbers are generally better lit and closer in; where two show the same thing, prefer the higher one.



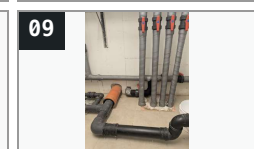
07



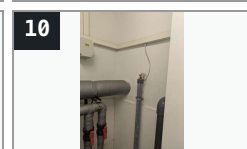
08



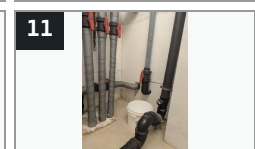
09



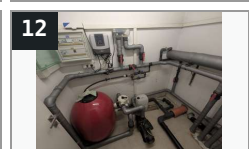
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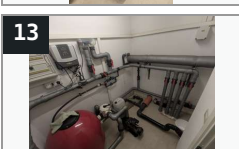
11



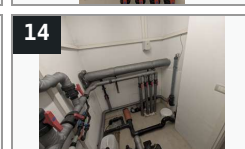
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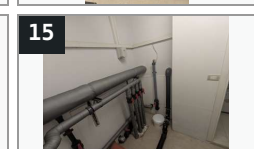
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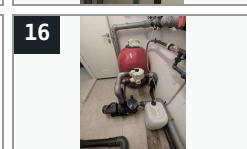
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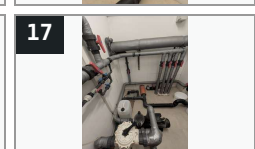
15



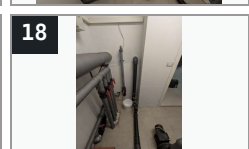
16



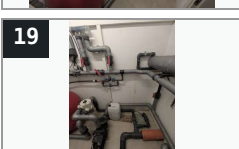
17



18



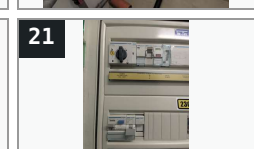
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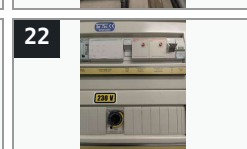
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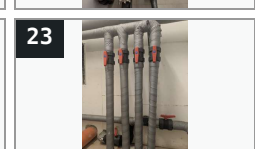
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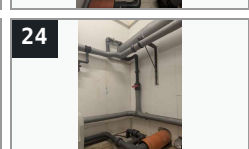
22



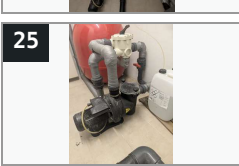
23



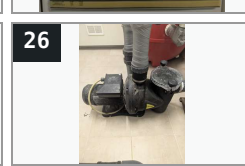
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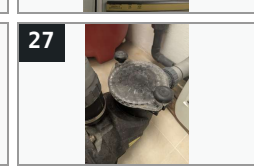
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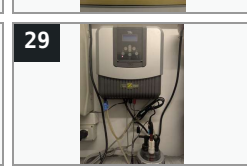
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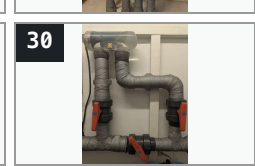
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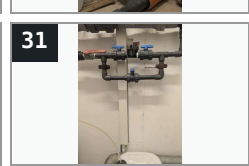
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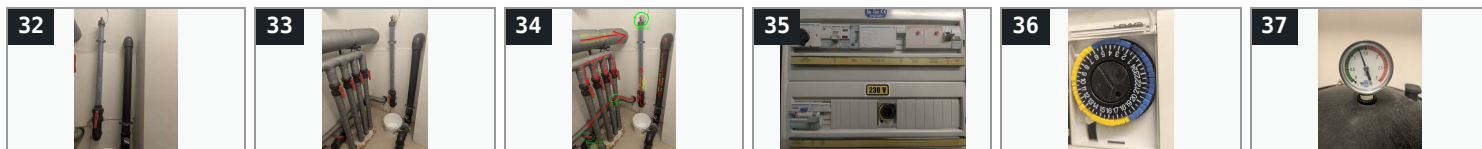


30



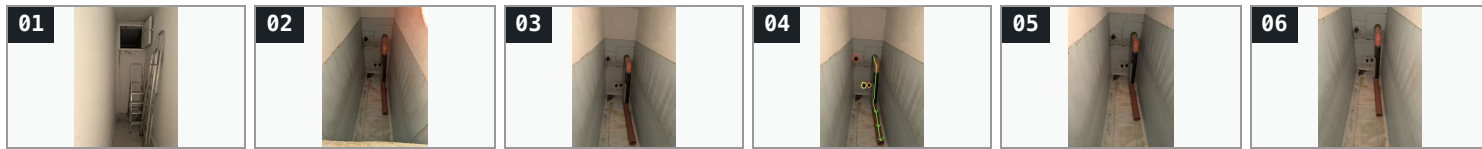
31





**05\_BALANCE-ROOM/** 6 photographs

The tank and the hatch that reaches it. The coloured marks on 04 are explained in that folder's ANNOTATION-KEY.txt.



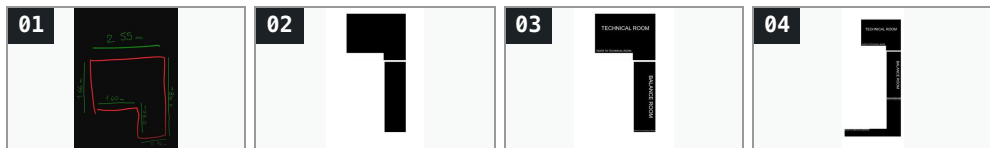
**06\_POOL/** 2 photographs

The pool itself, 52 seconds apart from opposite ends. Shape, layout and the overflow edge. Both are ultra-wide, an 18 mm equivalent by their own EXIF, so nothing here may be scaled or measured.



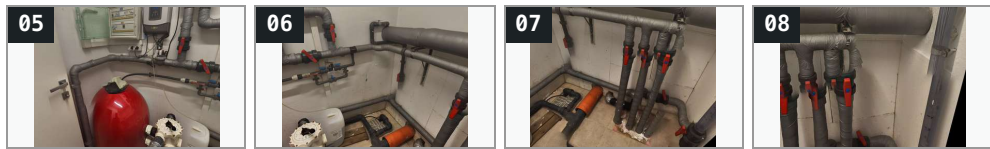
**07\_DRAWINGS/** 4 images

The owner's own drawings. 04 supersedes 02 and 03; its service corridor is topology only and must never be scaled.



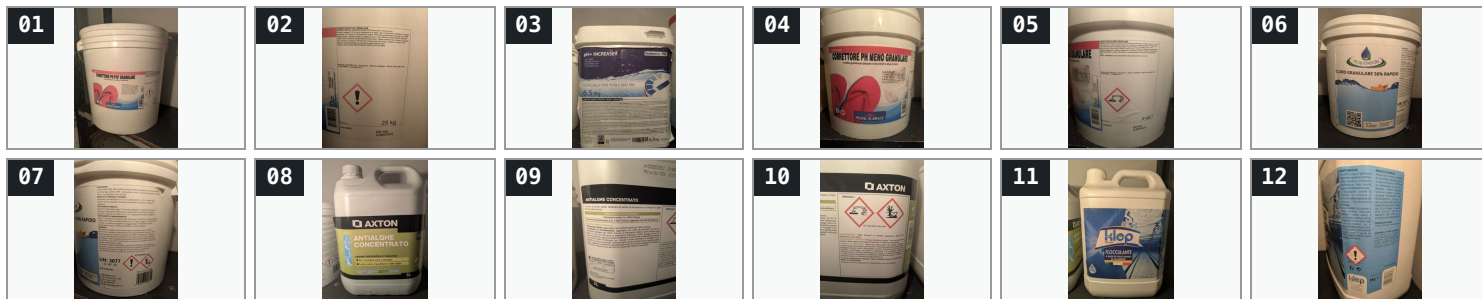
**09\_PANORAMAS/** 4 images

Straightened views reprojected from the single photosphere, 04\_plant-room/01. These are the wide images that can be counted from.



**11\_PRODUCTS/** 22 files, 21 of them shown here as pictures

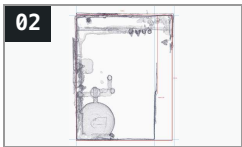
Every chemical on the shelf, each with its label. The working, the lot numbers and the hazard statements are in that folder's CATALOGUE.txt.







12\_SCAN/ 2 files, 1 of them shown here as a picture  
The 3D scan's derived plan. Good for geometry, useless for counting anything thin.



Not shown above, because a thumbnail would say nothing about them: 08\_video/ holds 3 videos, two sweeps of the room and one at floor level, and 10\_manuals/ holds 9 manuals, the makers' own documents with the official link for each in its SOURCES.txt.

**§14What is in the folder**

| FOLDER             | WHAT IT HOLDS                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00_inbox/          | Staging. New photographs and documents go in here named however they arrived, and are filed and renamed from there.                                                                                                                                                                                                                                 |
| 01_survey/         | This document as HTML and PDF, plus the diagram as a PNG. The HTML is entirely self-contained: no internet, no linked files, every photograph embedded in it.                                                                                                                                                                                       |
| 02_printed-sheets/ | The four printed pages kept in the room, photographed. Everything in the legend, the procedures and the water figures is transcribed and translated from these.                                                                                                                                                                                     |
| 03_nameplates/     | The five plates and labels every specification rests on: the Kripsol filter plate, the DAB pump nameplate, the valve dial, and the acid drum.                                                                                                                                                                                                       |
| 04_plant-room/     | Thirty-six photographs of this room, numbered in the order they were taken. Number 34 is the manifold wall with your own annotations on it; 35 to 37 are the night pass that reads the panel labels and the gauge needle.                                                                                                                           |
| 05_balance-room/   | Six photographs of the balance tank next door and the hatch that reaches it, with <i>ANNOTATION-KEY.txt</i> recording what the coloured marks on photograph 04 mean.                                                                                                                                                                                |
| 06_pool/           | The pool itself, from both ends. Shape, layout and the overflow edge. <b>Not dimensions:</b> both frames are ultra-wide, an 18 mm equivalent by their own EXIF, and a volume once read off them was a third low and withdrawn.                                                                                                                      |
| 07_drawings/       | Your own drawings: the dimensioned sketch of this room, and the scaled plan showing both rooms and how they sit together. The plan is genuinely to scale, checked at 296 pixels per metre against your measurements.                                                                                                                                |
| 08_video/          | Three phone pans: two 360 degree sweeps and one floor-level survey.                                                                                                                                                                                                                                                                                 |
| 09_panoramas/      | Four flat, straightened wall views, each reprojected from the 39-megapixel photosphere and <b>safe to count from:</b> photograph 07 has been checked at four drops. The numbering starts at 05 because 01 to 03 were stitched images that drew six drops where there are four, and were deleted. The folder's <i>READ-ME.txt</i> keeps that lesson. |
| 10_manuals/        | The makers' own manuals for the filter, the chlorinator, the TRi PRO module and the pump, with <i>SOURCES.txt</i> giving the official link for each and the figures taken from it.                                                                                                                                                                  |
| 11_products/       | The nine chemicals on the shelf, each photographed with its label, grouped by what they do. <i>CATALOGUE.txt</i> gives the active ingredient, the dose worked out                                                                                                                                                                                   |

| FOLDER   | WHAT IT HOLDS                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | for this pool, the hazards worst first, and a symptom-to-action guide.                                                                                                                                                                                                    |
| 12_scan/ | A 3D scan of this room, 1.29 million triangles. Its geometry independently confirms the tape measure: 2.52 m against 2.55 along the long axis, 2.00 against 1.96 across. Not usable for counting anything, because thin pipework does not survive a scan at that density. |

Measured dimensions for both rooms, and what follows from them, are in *DIMENSIONS.txt*. What every chemical is and what to reach for is in *11\_products/CATALOGUE.txt*. Every file is named to one rule, set out in *NAMING.txt* at the top of the folder: an index, then what the file shows, then an optional note on how it differs from its neighbour, then a marked technical token. For proof of what is installed, the 39-megapixel photosphere and the panel photograph are the record. This document is an interpretation of them.

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Compiled 2026-09-06 from photographs of the room and from the printed sheets kept in it. The sheets are a household reference rather than a professional specification, and do not necessarily capture everything about this installation. Nothing here has been verified by a physical inspection or against an installation drawing. For evidence of what is installed, the 39-megapixel photosphere and the panel photograph are the record.